SMART HEALTH RESPONSE APPLICATION FOR ASSISTING INJURED INDIVIDUALS

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PREFACE

This graduation project was developed to address the increasing need for rapid and effective medical assistance during emergencies occurring in public places. Throughout the project, our team aimed to design a smart and practical solution that helps reduce response time, improve communication between reporters and responders, and support injured individuals until professional help arrives.

Working on this project allowed us to explore modern technologies such as mobile application development, real-time location services, and smart emergency systems. We faced several challenges during the research, analysis, and design phases, but each challenge contributed to strengthening our understanding and improving the final outcome.

We would like to express our appreciation to our supervisors and everyone who supported us throughout this journey. Their guidance and feedback played an essential role in completing this work successfully.

We hope that this project contributes to enhancing community awareness, improving emergency response, and supporting efforts to save lives.

ACKNOWLEDGEMENT

We would like to extend our sincere gratitude and appreciation to everyone who contributed to the successful completion of this graduation project.

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Finally, we acknowledge that the completion of this work would not have been possible without the collective effort and dedication of every team member. It has been a privilege to collaborate on a project we hope will have a meaningful impact.

To all, we express our profound thanks.

Table of Content

PREFACE I

ACKNOWLEDGEMENT II

ABSTRACT IX

KEYWORDS: IX

1. INTRODUCTION 1

1.1. PROJECT OVERVIEW STATEMENT 1

1.2. GOALS & OBJECTIVES 2

1.3. PROJECT SCOPE 2

1.4. LIMITATIONS AND CONSTRAINTS 3

1.5. ASSUMPTIONS 3

2. LITERATURE REVIEW 5

2.1. PROJECT BACKGROUND 5

2.2. PREVIEWS WORK 6

2.3. SUMMARY OF PREVIOUS WORKS 10

2.4. RESEARCH GAP 10

3. FEASIBILITY STUDY 12

3.1. PURPOSE OF THE FEASIBILITY STUDY 12

3.2. JUSTIFICATION FOR THE PROPOSED SYSTEM 12

3.3. ECONOMIC FEASIBILITY 13

3.4. TECHNICAL FEASIBILITY 13

3.5. DESIRED SYSTEM FUNCTIONALITY 13

4. PROJECT PLAN 16

4.1. WORK BREAKDOWN STRUCTURE 16

4.2. ACTIVITY AND TASK LIST 17

4.3. GANTT CHART 18

5. SRS – SYSTEM REQUIREMENTS SPECIFICATION 19

5.1. DEVELOPMENT METHODOLOGY 19

5.2. SYSTEM REQUIREMENTS ANALYSIS 20

5.3. HARDWARE REQUIREMENTS SPECIFICATION 24

5.4. SOFTWARE REQUIREMENT SPECIFICATION 25

5.5. OTHER REQUIREMENTS SPECIFICATION 27

6. SYSTEM DESIGN 29

6.1. Applications Architecture Design 29
6.2. Models 31
.6.3 Database Design 48
.6.4 Database Table overview 49
6.5Audit Logs Table 50
6.6AEDs Table 51
6.7.Database Design 52
6.8. Facilities Table 53
6.9. Database Design 53
6.10.Reporters Table 54
6.11. Responders Table 54
6.12.Notifications Table 55

7. IMPLEMENTATION OF USER INTERFACES 56

7.1. Main Screen 57
7.2. Create Account Screen 58
7.3. Sign In Screen 59
7.4. Emergency Access Screen 60
7.5. Admin Dashboard Screen 63
7.6. Reporter Dashboard Screen 68
7.7. Responder Dashboard Screen 69
7.8. Receiving Facility Dashboard Screen 71

8. IMPLEMENTATION OF KEY SYSTEM MODULES 73

8.1. User Identity Verification Unit 73
8.2. Emergency Response Coordinator 73
8.3. Precise Geolocation and Routing Module 73
8.4. Medical Data and Records Management Module 73
8.5. Facility Resource Optimization Unit 73
8.6. Telemedicine and Multimedia Communications Unit 73

9. TESTING AND VALIDATION 75

9.1. Type of testing 75

10. CONCLUSION AND FUTURE WORK 78

10.1. Summary of Results 78
10.2. Limitations 79
10.3. Future Enhancements 80

REFERENCES **83**

List of Figures

FIGURE 2 1 THIS IS INTERFACE OF EUALERT APP 8

FIGURE 2-2 THIS IS INTERFACE OF ASAFNY APP 9

FIGURE 4-1 WORK BREAKDOWN STRUCTURE (WBS) 16

FIGURE 4-2 GANTT CHART 18

FIGURE **5-1** WATERFALL METHODOLOGY 19

FIGURE 5-2 FLUTTER 26

FIGURE 5-3 FIGMA 26

FIGURE 5-4 DRAW.IO 27

FIGURE 5-5 MS WORD 27

FIGURE 5-6 FIREBASE 28

FIGURE 6-1 THREE TIER ARCHITECTURE 29

FIGURE 6-2 USE CASE DIAGRAM FOR ADMIN 31

FIGURE 6-3 USE CASE DIAGRAM FOR REPORTER 32

FIGURE 6-4 USE CASE DIAGRAM FOR RESPONDER 32

FIGURE 6-5 USE CASE DIAGRAM FOR RECEIVING FACILITES 33

FIGURE 6-6 DFD LEVEL 0 (CONTEXT LEVEL 34

FIGURE 6-7 DFD-LEVEL-1 FOR ADMIN 35

FIGURE 6-8-DFD LEVEL 1 FOR REPORTER 35

FIGURE 6-9 DFD LEVEL 1 FOR RESPONDER 36

FIGURE 6-10 DFD-LEVEL-1 FOR RECEIVING FACILITIES 36

FIGURE 6-11 ACTIVITY DIAGRAM FOR ADMIN 37

FIGURE 6-12 ACTIVITY DIAGRAM FOR REPORTER 38

FIGURE 6-13 ACTIVITY DIAGRAM FOR RESPONDER 39

FIGURE 6-14 ACTIVITY DIAGRAM FOR RECEIVING FACILITY 40

FIGURE 6-15 SEQUENCE DIAGRAM FOR ADMIN 42

FIGURE 6-16 SEQUENCE DIAGRAM FOR REPORTER 43

FIGURE 6-17 SEQUENCE DIAGRAM FOR RESPONDER 44

FIGURE 6-18 SEQUENCE DIAGRAM FOR RECEIVING FACILITIES 45
FIGURE 6-19 CLASS DIAGRAM 46
FIGURE 6-20 ENTITY RELATIONSHIP DIAGRAM (ERD) 48
FIGURE 7-1 MAIN SCREEN 57
FIGURE 7-2 CREATE ACCOUNT SCREEN 58
FIGURE 7-3 SIGN IN SCREEN 59
FIGURE 7- 4 EMERGENCY ACESS SCREEN 60
FIGURE 7- 5 EMERGENCY STATUS SCREEN 61
FIGURE 7- 6 SELECT ROLE SCREEN 62
FIGURE 7- 7 ADMIN DASHBOARD SCREEN 63
FIGURE 7- 8 MANAGE USERS SCREEN 64
FIGURE 7- 9 RESPONDER APPROVAL SCREEN 65
FIGURE 7- 10 MANAGE CASES SCREEN 66
FIGURE 7- 11 EMERGENCY ALERTS SCREEN 67
FIGURE 7- 12 REPORTER DASHBOARD SCREEN 68
FIGURE 7- 13 RESPONDER DASHBOARD SCREEN 69
FIGURE 7- 14 CASE MANAGMENT SCREEN 70
FIGURE 7- 15 RECEIVING FACILITY DASHBOARD SCREEN 71
FIGURE 7- 16 CASE DETAILS SCREEN 72
FIGURE 8-1 THIS IS FIGURE 2 ..... **ERROR! BOOKMARK NOT DEFINED.**
FIGURE 10-1 THIS IS FIGURE 2 ..... **ERROR! BOOKMARK NOT DEFINED.**
FIGURE 11-1 THIS IS FIGURE 2 ..... **ERROR! BOOKMARK NOT DEFINED.**

LIST OF TABLES

TABLE 4-1 ACTIVITY AND TASK LIST 17

TABLE 7-1 THIS IS THE CAPTION FOR TABLE ..... **ERROR! BOOKMARK NOT DEFINED.**

TABLE 8-1 THIS IS THE CAPTION FOR TABLE ..... **ERROR! BOOKMARK NOT DEFINED.**

TABLE 10-1 THIS IS THE CAPTION FOR TABLE 77

TABLE 11-1 THIS IS THE CAPTION FOR TABLE ..... **ERROR! BOOKMARK NOT DEFINED.**

ABSTRACT

In modern societies, many health emergencies occur unexpectedly in public places such as markets, universities, and parks. The time between the occurrence of the situation and the arrival of critical medical assistance is often the difference between life and death. Delays caused by traffic, a lack of immediate communication, or difficulty locating the incident can exacerbate the situation. Therefore, utilizing smart technologies and increasing public awareness about first aid can significantly reduce response times and save lives. Hence, the idea of the graduation project " Smart Health Response Application for Assisting Injured Individuals" to be a smart link between people at the scene of the emergency (informants, witnesses), health professionals, volunteers, and even official authorities such as the ambulance and the Red Crescent. The app allows anyone to report a case with the push of a button, and the system immediately locates the location, publishes alerts to those nearby, and provides immediate ambulatory instructions until specialized help arrives. By harnessing technology and community awareness, the app aims to maximize the chances of survival and minimize the risk of delay.

Keywords:
Health Response, Injury Assistance, Injured individuals, Emergency Response, Smart Health Application, First Aid.

1. INTRODUCTION

These days, we are witnessing a noticeable and alarming increase in the number of accidents and injuries happening in different public places. Unfortunately, in many cases, the main reason that people become seriously ill, suffer long-term complications, or even lose their lives is not the injury itself, but rather the delayed response to the emergency. The longer it takes for the victim to receive proper medical assistance, the higher the chances of permanent damage or death.

This highlights a critical gap in our current emergency response systems and calls for innovative solutions that can save lives by reducing response time.

Therefore, our project aims to design and develop a smart mobile application that provides rapid medical assistance during emergencies that occur in public spaces. The application is intended to serve as a reliable bridge between witnesses or informants who report an incident and the professionals or volunteers who are capable of providing medical support. It will also integrate with official agencies such as ambulances, hospitals, and humanitarian organizations like the Red Crescent to ensure that professional medical help is dispatched as quickly as possible.

1.1. Project Overview Statement

The project aims to provide rapid medical response in emergency situations through a mobile application that connects witnesses, informants, specialists, volunteers, and official agencies (such as ambulances and the Red Crescent), providing first aid instructions until help arrives, thus reducing delays and increasing the chances of survival.

Exiting System

There is currently no single comprehensive system that can provide a rapid medical response through a mobile application that connects witnesses, informants, specialists, and volunteers and links them directly to the ambulance.

Proposed System

The proposed system is a mobile application designed to provide rapid medical response during emergencies in public places the App provide the following features:

- Allows instant reporting of incidents with one click.
- Automatically detects and shares the location of the case.
- Send alerts to nearby health professionals, volunteers, and official agencies (ambulance, Red Crescent).
- Provides immediate first aid to witnesses until help arrives.
- Reduces response time and increases survival chances.

1.2. Goals & Objectives

Goals:

In this project, we aim to accelerate the response to health emergencies in public places by providing a smart app that connects informants, responders, and medical authorities in real time, saving lives and minimizing the risk of delays.

Objectives:

- Create a community health support network that includes specialists, volunteers, and citizens for reporting emergency health cases.
- Enable non-specialized individuals to provide correct first aid under direct medical supervision.
- Reduce the initial response time in medical emergencies.
- Raising health and ambulatory awareness among community members.
- Improving the chances of surviving medical emergencies outside of health facilities
- To enhance coordination between healthcare authorities and emergency medical service providers.

1.3. Project Scope

The scope of this project is to develop a mobile application aimed at facilitating a collective health response for individuals affected by injuries or medical emergencies within the Jazan

region. The application will serve as a centralized digital platform that enables users to report emergency health incidents in real-time, share their location with healthcare responders, and receive immediate assistance.

1.4. Limitations and Constraints

- Internet Connectivity:
  The application requires a stable internet connection to transmit emergency alerts and share real-time locations. In areas with weak or no network coverage, the system performance may be limited.
- Device Compatibility:
  The app depends on smartphones (Android and iOS). Users without compatible devices may not benefit from the system.
- Battery and Power Dependency:
  Continuous use of GPS and notifications consumes battery power, which may affect the app's availability during emergencies.
- User Training and Awareness:
  Non-specialized individuals may face difficulties in providing accurate information or using the app effectively without basic awareness and guidance.
- Data Privacy and Security:
  The system must handle sensitive user data (location, medical emergencies) securely. Any breaches could compromise privacy.
- Dependence on External Responders:
  The app can notify medical authorities and volunteers, but the actual response time still depends on the availability and readiness of emergency services.

1.5. Assumptions

This application is designed to support and rescue injured or ill individuals during emergencies by enabling quick and effective incident reporting and response. To achieve this goal, the project is based on a set of fundamental assumptions related to the availability of technology, communication infrastructure, and community readiness, all of which are essential to ensure the effective operation and optimal utilization of the application in the Jazan region.

- We assume that the application will help to reduce response time and increase survival chances.
- We assume that reliable internet connection is available in all targeted areas.
- We assume that users have smartphones capable of running the application.

- We assume that any individual at the emergency site—whether the injured person, a first responder, or a reporter/bystander—can submit an emergency report through the application.
- We assume that users allow the application to access their GPS location to accurately identify the emergency site.
- We assume that emergency response teams or trained volunteers are available to promptly handle the reports.
- We assume that reports include clear and accurate information about the injured person or the incident to accelerate the response process.

2. LITERATURE REVIEW

2.1. Project Background

In recent years, road accidents have become a serious issue, often resulting in fatalities due to delays in providing medical attention. The Metropolitan Manila Accidents Reporting and Analysis System (MMARAS) reported more than 109,000 accidents in 2016, with hundreds of fatalities and thousands of injuries. Most victims rely on bystanders to reach the hospital, which often leads to late response.

The rise of Internet of Things (IoT) technologies and mobile applications has enabled solutions to improve emergency response. Witnesses can use smartphones to report accidents, share real-time GPS location, and communicate with rescuers through interactive chat. By providing first aid guidance and connecting victims with the nearest responders, survival chances can significantly increase.

This project aims to address these issues by developing SAFE, a smart mobile application that helps users request immediate medical assistance, share location, and ensure rapid communication between reporters and medical responders [1]

Smartphone applications have become critical tools in emergency medical services (EMS), enhancing response time and delivering immediate care in emergency situations. These mobile health (mHealth) solutions include features such as real-time location tracking, incident reporting, and first aid guidance. Effective communication between users and EMS personnel is vital, yet challenges related to data reliability, communication security, user usability, and adaptability to diverse settings persist. This study highlights the role of digital health interventions in strengthening emergency response systems by improving alert accuracy and reducing response delays, thus contributing to overall EMS effectiveness [2]

Delays in prehospital care are frequently associated with miscommunication during the call-taking process, particularly in providing accurate caller location. Such delays may compromise the effectiveness of emergency medical services. Digital technologies, such as GPS, have demonstrated efficiency in locating callers, thereby reducing response times. In Brazil, for instance, the SAMU system primarily depends on verbal location reporting, which is often time-consuming and susceptible to errors. To overcome this limitation, a mobile application was developed using design thinking methodologies to automatically transmit

the caller's geolocation to emergency services. This study assesses the app's usability and compares its performance with traditional emergency call systems [3]

2.2. Previews Work

An Interactive Mobile Application to Request the Help of the Nearest First Aider by the Injured (Areej Alshutayria, Nahla Aljojo, etc., 2021)

The study addresses the high Ambulance Response Time (ART) in Saudi Arabia, which exceeds the global standard of six minutes due to factors such as traffic congestion, difficult geographic locations, limited ambulance resources, and increasing population and pilgrim numbers. To tackle this, the proposed My Nearest Aider application aims to connect injured individuals or bystanders with the nearest qualified first aiders to provide immediate assistance while waiting for the ambulance. Unlike the existing ASAFNY app, which faced issues such as low public awareness, technical problems, and a complex interface, the proposed system focuses on faster response, easier communication, and improved accessibility. Future recommendations include supporting multiple languages, integrating secure phone verification, enabling live communication between users and first aiders, expanding usability for people with disabilities, and refining the interface design based on user feedback to ensure a more efficient and user-friendly emergency response system. [4]

Disadvantages:

-Low Public Awareness – A significant portion of potential users may not

be aware of the application, which reduces its overall impact during emergencies.

-Technical and Interface Challenges – Bugs, operational issues, or a

complex user interface can slow down response times and hinder usability in critical situations

-Limited Number of Certified First Aiders – The availability of trained

responders may be insufficient in certain regions, which can delay first aid delivery

SAFE: Smart Application for Emergency (John Nicolo, T.Curz, Gabrielle Angelo, etc., 2021)

The SAFE application is an Android-based emergency tool designed to facilitate and accelerate the reporting of accidents requiring urgent medical attention. It allows users to request immediate help from registered medical responders with features such as live location

sharing and real-time chat, reducing response time and enabling rescue teams to quickly reach victims compared to traditional methods like hotlines. Study results showed that the application enhances communication efficiency and increases the chances of saving lives, while researchers suggested improvements such as categorizing emergency types and specifying the required responders for each case to further improve user experience and response effectiveness.[1]

Disadvantages:

-The app is currently only available in the Philippines, limiting its use for international or out-of-region users.

-Requires a stable internet connection and GPS, which may not work in remote or poorly covered areas

-Currently, the app may only support local languages, restricting accessibility for non-Filipino users or tourists

EUalert app

EU Alert! is an app developed by Emergency University aimed at large organizations such as corporations and government agencies. It is part of corporate emergency systems that allow ordinary individuals to report medical emergencies such as sudden cardiac arrest, as well as non-medical emergencies such as fires or earthquakes. The app alerts trained responders within the organization, providing them with accurate information about the location of the incident and the whereabouts of necessary equipment such as AEDs.

EU Alert! can accelerate response within an organization by connecting reports directly to nearby responders, as well as educating and training workers by simulating different emergencies. [5]

Disadvantages:

- The app is only available to customers associated with Emergency University, limiting its reach

- Its success is also highly dependent on the availability of trained responders and medical equipment within the organization

-issues of privacy and data protection, location accuracy, and the institutional cost of implementing the system and training staff remain challenges that may affect its effectiveness



Figure 2 1 This is interface of EUalert app

Asafny app

Asafni is one of the digital solutions launched by the Saudi Red Crescent Authority in 2018 to enhance emergency and ambulance services in the Kingdom. The app aims to facilitate the process of uploading ambulance reports directly to the operating rooms, with the option to accurately geolocate and send basic medical data about the user such as chronic diseases and medications used , The application is also characterized by its support for people with special needs such as the deaf and hard of hearing, as it allows the possibility of sending reports without the need for voice communication, in addition to providing a multilingual interface to meet the needs of diverse populations. [6]

Disadvantages:

- technical challenges such as difficulty registering at times or the app failing to confirm codes,

-issues with location accuracy or connection stability in some regions.

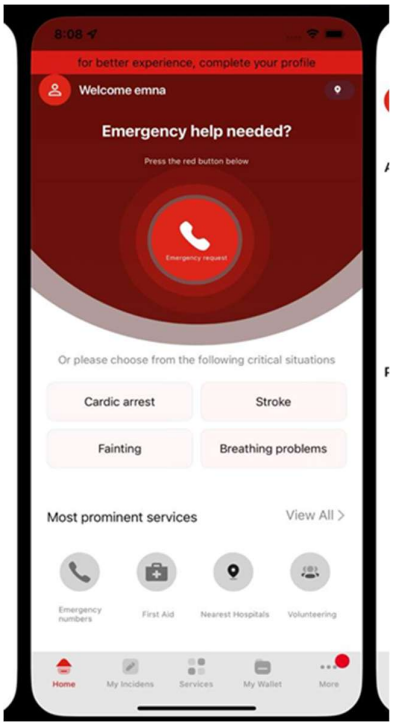


Figure 2-2 This is interface of Asafny app

2.3. Summary of Previous Works

Name of app:	Asafny	EUalert	SAFE
Main Goals:	Facilitating the submission of medical emergency reports in the Kingdom and linking them to the Saudi Red Crescent, while supporting people with special needs.	Enabling organizations to manage emergencies within buildings and connect trained responders to the incident immediately.	Provide a mobile tool for reporting medical emergencies, enabling quick communication, location sharing, and first aid guidance.
Scope of use:	General within the Kingdom of Saudi Arabia for all citizens and residents	Institutional for institutional clients only	Android users in the Philippines for medical emergencies.
Advantage:	Support for people with special needs, multilingual, documenting reports, available to citizens and residents	Rapid response within the organization, training and awareness of responders, covering multiple emergencies	Fast communication, GPS location, user-friendly interface.
Disadvantage:	Technical problems in registering or updating websites, depends on the Internet connection	Dedicated to institutional clients only, Depends on availability of trained responders, Accurate location within buildings	Android only, Philippines only, requires internet, relies on user knowledge.

Table 2 1 This is the summary of emergency response application

This table provides a summary of applications like ours, highlighting the advantages and disadvantages of each.

2.4. Research Gap

Our app:

provides immediate and accurate response to medical emergencies, accessible to everyone through an easy-to-use interface while ensuring user data protection. It offers quick and effective emergency response, eliminating the need to rely on specific responders, although improvements are needed to address the current reliance on limited responders within certain environments

3. FEASIBILITY STUDY

A feasibility study is a preliminary systematic evaluation aimed at determining the viability and potential success of a proposed project or new business venture. It involves analyzing risks, benefits, costs, challenges, timelines, budgeting, market research, supply chain factors, technological considerations, and legal requirements to enable stakeholders to make informed decisions on whether to pursue, modify, or abandon the project. The study typically concludes with a feasibility report that presents the findings and outlines the overall viability and implementability of the project. [7].

3.1. Purpose of the Feasibility Study

The purpose of this feasibility study is to determine whether the proposed system—Smart Health Response Application for Assisting Injured Individuals, a mobile application designed to provide rapid medical response during emergencies in public places—is practical and achievable. The study aims to evaluate the project from different aspects, such as technical feasibility, operational feasibility, and overall usability, in order to understand how easy and effective the system will be to implement and use.

3.2. Justification for the proposed system

1. **Operational Purpose:**

   To create an integrated system that reduces the time between the moment of emergency and the arrival of life-saving medical care to the injured person.

2. **Marketing Purpose:**

   Strategic partnership with official authorities: Cooperation with the Ministry of Health, the Red Crescent lends enormous credibility to the application and facilitates it to a wide segment of the public through the channels of these entities.

3. **Scalability and Sustainability:**

   Geographical Expansion: The project can be started in one city as a pilot phase, then easily expand to other cities and regions, and finally cover the entire country or even other Arab countries.

3.3. Economic Feasibility

Economic feasibility is the assessment of whether a proposed project or investment is financially viable. It involves analyzing the costs, revenues, potential savings, and long-term financial impacts to determine if the expected benefits outweigh the expenses. This analysis helps decision-makers evaluate the profitability, sustainability, and overall economic justification of a project before committing resources.[8]

The development and implementation of our system will involve costs related to hardware, software development, server hosting, and maintenance. Utilizing open-source technologies and tools like Flutter for app development, Firebase for backend services, and Figma for UI/UX design can significantly reduce the software costs. So, our project is feasible economically.

3.4. Technical Feasibility

- **Definition:**
  - Technical feasibility evaluates if the project can be developed and
  - implemented with available technology, resources, and expertise.
- **Technology Required:**
  - Frontend: Flutter (cross-platform mobile development, responsive
  - UI, hot-reload)
  - Backend: Firebase (real-time database, user authentication, cloud
  - messaging, encrypted data)
  - Design Tools: Figma (UI/UX design, wireframes, prototyping)
- **Hardware Requirements:**
  - Android and iOS smartphones
  - Internet connectivity (for real-time updates and notifications)
  - Firebase cloud services (no physical servers needed)
- **Feasibility Assessment:**
  - All tools and platforms are available and accessible
  - Flutter supports cross-platform development
  - Firebase provides efficient backend and real-time communication
  - Figma allows collaborative and consistent UI design

3.5. Desired System Functionality

The key functionalities of the application will include:

- **User Registration and Authentication:**

  Allow users to securely create accounts, log in, and manage their profiles within the system.

- **Emergency Assistance Request**:

  Enable users to request medical assistance anytime and anywhere with automatic location detection to ensure accurate response.

- **Volunteer Response and Navigation:**

  Allow volunteers to view nearby emergency cases and navigate directly to the injured person using the integrated map feature.

- **First Aid Guidance:**

  Provide clear and interactive first aid instructions through videos, audio clips, and images to assist volunteers without prior medical training.

- **Personal Health Profile:**

  Enable users to create and maintain a health profile that includes vital medical information such as blood type, chronic conditions, medications, and emergency contacts.

- **Enhanced Response Time:**

  Reduce response time by providing instant alerts and accurate location sharing between users and volunteers.

- **Public Awareness and Education:**

  Offer educational content (videos, texts, and audio materials) to raise public awareness about first aid and emergency procedures.

- **User-Friendly Interface:**

  Provide a simple, intuitive, and accessible interface designed for quick interaction during emergency situations.

- **Administrative Management and Reporting:**

Allow the administrator to monitor system activities, manage users and volunteers, document incidents, and generate analytical reports to evaluate and improve system performance.

4. PROJECT PLAN

Project planning is the process of defining the goals, timeline, tasks, resources and strategy for executing a project. It outlines what needs to be done, when and by whom to achieve the project's goals. Anyone involved in the project, from the project manager to team members, can reference the project plan at any stage of the project to better understand their roles and responsibilities. [9]

4.1. Work Breakdown Structure

A work breakdown structure (WBS) is a visual, hierarchical and deliverable-oriented deconstruction of a project. It is a helpful diagram for project managers because it allows them to break down their project scope and visualize all the tasks required to complete their projects.

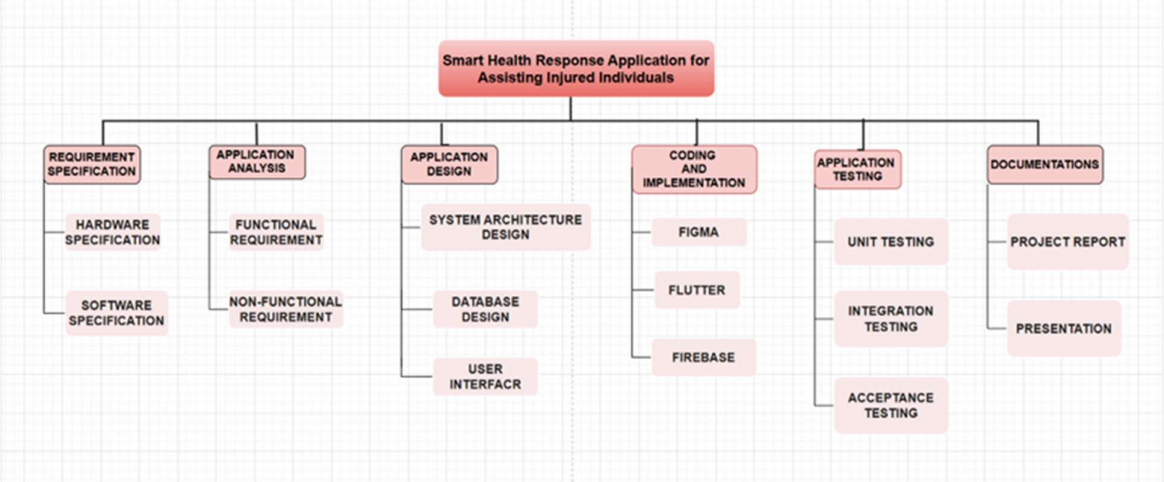


Figure 4-1 Work Breakdown Structure (WBS)

This diagram illustrates the project phases and the tasks of each phase.

4.2. Activity and Task List

An activity list is a document that outlines a set of organized activities or tasks related to a specific goal or area. It serves to clarify what needs to be done or what has already been completed. Activity lists are useful for planning, tracking progress and managing time and effort in various contexts such as work, education or personal projects.

Phase	Task	Duration	Start Date	End Date
Smart Health Response Application for Assisting Injured Individuals		60	7/9/25	27/11/25
		12	7/9/25	22/9/25
Requirements Specification	Hardware Specification	6	7/9/25	14/9/25
	Software Specification	6	15/9/25	22/9/25
		18	23/9/25	16/10/25
System Analysis	Functional Requirement	9	23/9/25	5/10/25
	Non-functional Requirement	9	6/10/25	16/10/25
		30	19/10/25	27/11/25
System Design	System Architecture Design	6	19/10/25	26/10/25
	Database Design	9	27/10/25	6/11/25
	User Interface	15	9/11/25	27/11/25

Table 4-1 Activity and Task list

Provides a breakdown of the project into main phases (Requirements Specification, System Analysis, System Design), listing tasks, durations (in days), and start/end dates.

4.3. Gantt Chart

A Gantt Chart serves as a project management tool designed to visually outline a project's schedule. It organizes tasks or activities along the vertical axis, while time is displayed on the horizontal axis. Horizontal bars represent each task, illustrating their start date, end date, and duration.

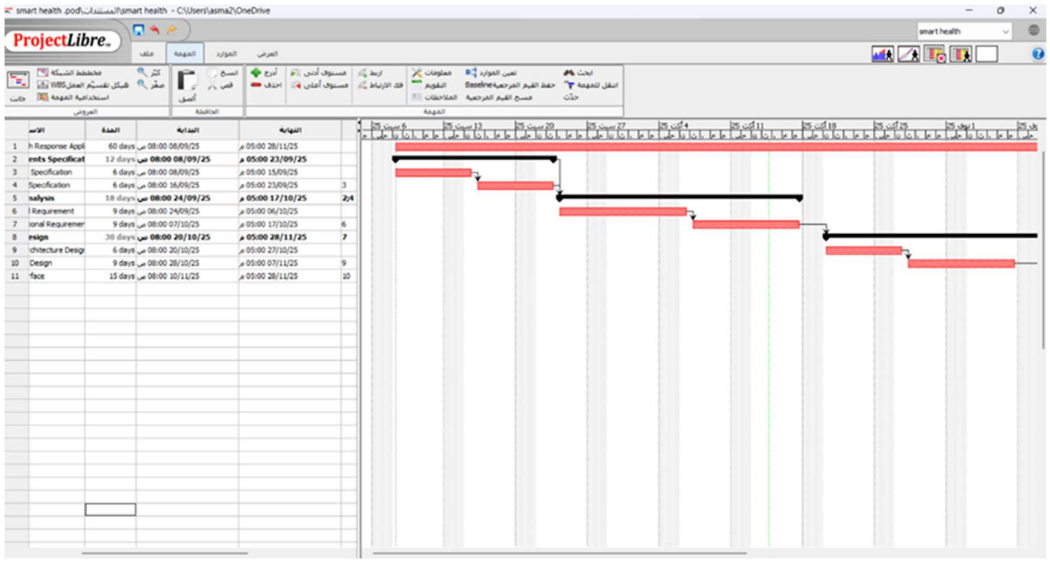


Figure 4-2 *Gantt Chart*

A visual representation of the project schedule, illustrating the timeline for tasks in horizontal bars using the ProjectLibre tool.

5. SRS – SYSTEM REQUIREMENTS SPECIFICATION

The Software Requirement Specification (SRS) format serves as a comprehensive outline and detailed explanation of the requirements needed for the software's successful development. These requirements encompass both functional and non-functional aspects, based on the specific needs of the project. Effective communication between customers and contractors is essential to ensure a clear understanding of customer expectations and necessities.[10]

5.1. Development Methodology

A software development methodology is a framework used to structure, plan, and manage the software development process. Its goal is to ensure the delivery of software in accordance with project specifications, within time and budget, and with minimized project risks.[11]

5.1.1 Waterfall Methodology

The Waterfall methodology is a linear and sequential project management approach in which each phase, from analysis and design to development and testing, is completed before the next phase begins. This methodology emphasizes careful upfront planning, detailed documentation, and structured execution to ensure clear requirements, predictable timelines, and measurable progress.[12]

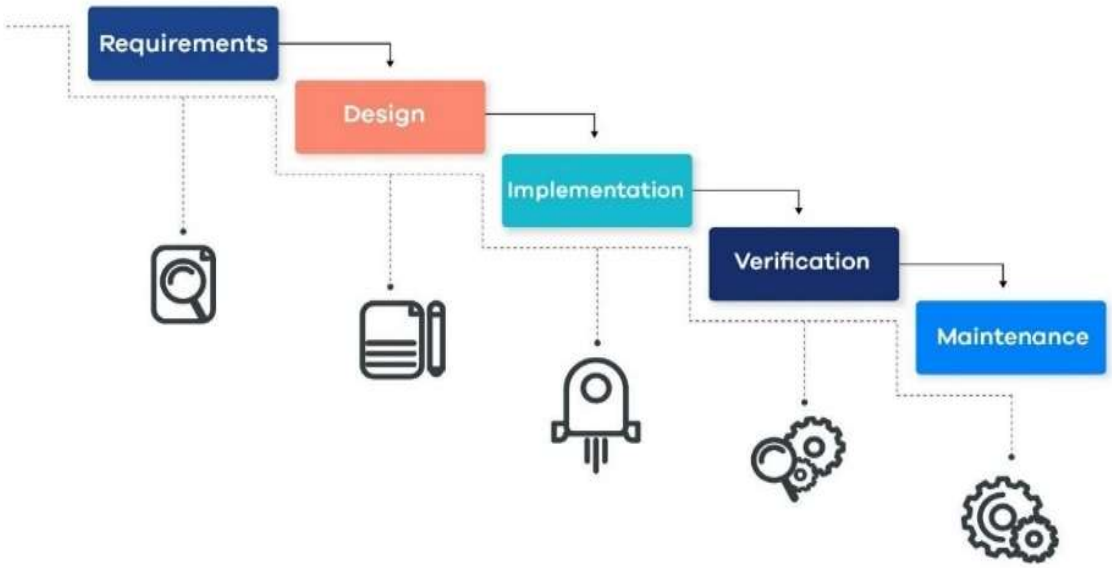


*Figure **5-1** Waterfall Methodology*

Waterfall Methodology Phases

- **Requirements:**

  Identifying and understanding all project requirements upfront to create a clear document including costs, risks, assumptions, and timelines.

- **Design:**

  Creating a technical design for the project, including high-level structure and detailed low-level components.

- **Implementation:**

  Writing the code and implementing the application based on the design and requirements, with some initial testing.

- **Verification:**

  Testing the product to ensure it is error-free and all requirements are met.

- **Maintenance:**

  Releasing the product to the end user and performing updates and maintenance based on feedback and change requests.[12]

5.2. System Requirements Analysis

Application Requirements Analysis is a crucial phase in software development. It involves identifying, documenting, and analyzing all the functional and non-functional requirements needed for the mobile application to operate effectively and meet user needs.

5.2.1 Functional Requirements:

- **Admin:** A system administrator who manages users, monitors reports, and maintains overall system performance.
  - **Login:** Access the secure web-based admin control panel.
  - **Manage Users:** Add, edit, or deactivate reporters, responders, and facilities.
  - **Approve Responders:** Verify and activate responders after reviewing certifications.

- **Monitor System Activity:** View active incidents on a live Jazan map.
- **Manage Facilities & AEDs:** Update hospitals, clinics, and AED device locations.
- **System Maintenance:** Monitor server health and data synchronization.
- **Data Management:** Manage backups, data retention, and restoration.
- **Generate & Schedule Reports:** Create and send automated analytical reports.
- **Export & Share Data:** Download reports in PDF or Excel format.
- **Audit Logs:** Review all system actions and user activities.
- **Broadcast Emergency Alerts:** Send urgent notifications to responders in specific regions.

- **Reporter:** A user who reports medical emergencies through the application by sending alerts with location details.
  - **Registration:** Create an account using a phone number or email.
  - **Login:** Access the emergency dashboard after authentication.
  - **Send Emergency Alert (SOS):** Report emergencies instantly with GPS location and case type.
  - **Voice-Activated SOS:** Trigger an alert by voice command when unable to use the phone.
  - **Attach Media:** Upload photos or voice notes to describe the situation.
  - **Offline First Aid Guidance:** Follow CPR or first-aid steps without an internet connection.
  - **Share Live Location:** Allow continuous tracking of the user's real-time position.
  - **Emergency Contact Alert:** Automatically notify a saved contact once an alert is sent.

- **View ETA:** Display the responder's estimated arrival time on the map.
- **In-App Communication:** Chat or call securely with the responder or dispatcher.
- **Track Report Status:** Monitor case progress from received to closed.
- **Rate Service:** Evaluate and provide feedback after the case is completed.
- **Logout**: Securely exit their account.

- **Responder:** A trained individual who provides first aid and responds to emergency alerts in the field.
  - **Registration:** Register as a responder and upload CPR or first-aid certificates.
  - **Login:** Access the responder dashboard and manage availability.
  - **Receive Nearby Alerts:** Get instant notifications for nearby emergencies.
  - **Accept or Reject Alert:** Respond to or decline cases with a given reason.
  - **Smart Assignment:** Automatically assign cases based on distance and qualification.
  - **Navigation & Live Location:** Use map navigation while sharing real-time tracking.
  - **Medical Protocol Guidance:** View case-based instructions (CPR, bleeding, choking, etc.).
  - **In-App Communication:** Communicate with reporter and hospital during the case.
  - **Handover Report:** Complete a short form when transferring the patient.
  - **Close Case:** Submit the final report and mark the case as completed.
  - **Logout**: Securely exit their account.

- **Receiving Facility:** A hospital or medical center that receives patients and confirms case completion in the system.
    - **Login:** Access the facility dashboard to view active and incoming cases.
    - **View Incoming Alerts:** Receive notifications for patients being transferred.
    - **Track Responder Location:** Monitor live route and estimated arrival time (ETA).
    - **Communicate with Responder:** Exchange quick updates such as "ER ready" or "Use Gate 2."
    - **Confirm Case Reception:** Verify that the patient has arrived at the facility.
    - **Fill Handover Form:** Record patient details and receiving staff information.
    - **Update Case Status:** Change status to "Received," "In Treatment," or "Closed."
    - **Generate Internal Reports:** View statistics of received cases by date or type.
    - **Feedback to System:** Provide outcome notes to improve response quality.

5.2.2 Non-Functional Requirements:

- Performance:
    - Response Time: Send and process alerts within 3 seconds.
    - Network Efficiency: Operate smoothly even with weak connections.

- Security:
    - Data Encryption: Protect user and medical data at all times.
    - User Authentication: Apply strong authentication measures.
    - Access Control: Limit data access based on user roles.

- Reliability:
  - High Availability: Maintain 99.9% uptime for continuous service.
  - Error Recovery: Automatically recover from failures or interruptions.

- Usability:
  - Ease of Use: Provide a simple and intuitive interface for quick actions.
  - Accessibility: Ensure clear visuals and easy navigation under pressure.

- Availability:
  - 24/7 Operation: Ensure access to the system at any time.

- Maintainability:
  - Modular Design: Allow easy updates, fixes, and feature upgrades.
  - Error Logs: Record system events for quick troubleshooting.
  - Continuous Updates: Enable improvements without service disruption.

- Compatibility:
  - Cross-Platform: Support Android and iOS devices seamlessly.
  - System Integration: Connect with hospitals and emergency systems.

5.3. Hardware Requirements Specification

Item	Description

Phone (Client)	Latest flagship device (e.g., Samsung Galaxy S-series, Apple iPhone Pro)
Processor	3.2 GHz or higher
RAM	4 GB or higher
Hard Disk	32GB or higher
OS	Android 14 or higher, or iOS 17 or higher

Table 5-1 Hardware Requirement Specifications

5.4. Software Requirement Specification

Item	Description
Flutter	Frontend (Cross-platform mobile development with responsive UI and hot-reload)
Figma	Design Tools: (UI/UX design, wireframes, prototyping)
Draw.io	Program for creating UML diagrams
MS Word	For writing reports
Firebase	Backend (real-time database, user authentication, cloud messaging, encrypted data)

Table 5-2 Software Requirements Specifications

Flutter:

Open-source framework from Google for building mobile, desktop, and web apps by writing code just once.

Key features:

Beautiful and fast user interface: it uses a technology called "Skia" to make apps smooth and engaging

Language used: Dart.



Figure 5-2 Flutter

Figma:

A cloud-based (browser-based) tool for designing user interfaces (UIs), user experiences (UXs), and creating interactive prototypes.

Key Features:

Real-Time Collaboration: Multiple designers work on the same file simultaneously (like Google Docs).

No Installation Required: Works directly from the browser.

Interactive Prototyping: You can link screens together to preview how your app or website will work before coding.



Figure 5-3 Figma

Draw.io:

Is a free, web-based diagramming tool used for creating flowcharts, network diagrams, organizational charts, and other types of visual representations. It allows users to create and edit diagrams using drag-and-drop features, with various shapes, connectors, and templates. Draw.io supports collaboration, integrates with cloud storage services like Google Drive, and is commonly used for both personal and professional diagramming needs.



Figure 5-4 Draw.io

MS Word:

Microsoft Word (MS Word) is a word processing software developed by Microsoft. It is used for creating, editing, formatting, and printing text documents. MS Word offers a wide range of features such as text formatting, spell check, templates, images, tables, and collaboration tools, making it suitable for writing reports, letters, essays, and other types of documents.



Figure 5-5 MS word

5.5. Other Requirements Specification

Database

A **database** is an organized and structured collection of related and structured information, or data, that is electronically stored and managed by a Database Management System (DBMS). It enables the efficient storage, retrieval, modification, and control of data—including text, numbers, images, and videos—while maintaining integrity, security, and allowing concurrent access by multiple users or applications. Databases are fundamental to modern information systems, supporting a wide range of tasks such as managing customer transactions, product inventory, financial records, and large-scale enterprise analytics.

Firebase

Firebase is a comprehensive Backend-as-a-Service (BaaS) platform developed by Google that provides a wide range of cloud-based tools and services designed to help developers build, deploy, and manage high-quality web and mobile applications efficiently. It eliminates the need for developers to create and maintain backend infrastructure from scratch, allowing them to focus on application logic and user experience and widely used because of its ease of integration, real-time synchronization, cross-platform support, and tight integration with Google Cloud services. It supports frameworks like React, Angular, Flutter, and Android SDK, making it suitable for modern application development.



Figure 5-6 Firebase

6. SYSTEM DESIGN

6.1. Applications Architecture Design

Application Architecture Design is the fundamental blueprint that defines the structure and direction for developing and maintaining applications within an organization. It ensures alignment with business objectives, efficiency, and reliability while considering factors such as scalability, maintainability, and performance to drive business success.[13]

6.1.1 Three Tier Architecture

Three-tier architecture is a well-established software application architecture that organizes applications into three logical and physical computing tiers: the presentation tier, or user interface; the application tier, where data is processed; and the data tier, where application data is stored and managed.[14]

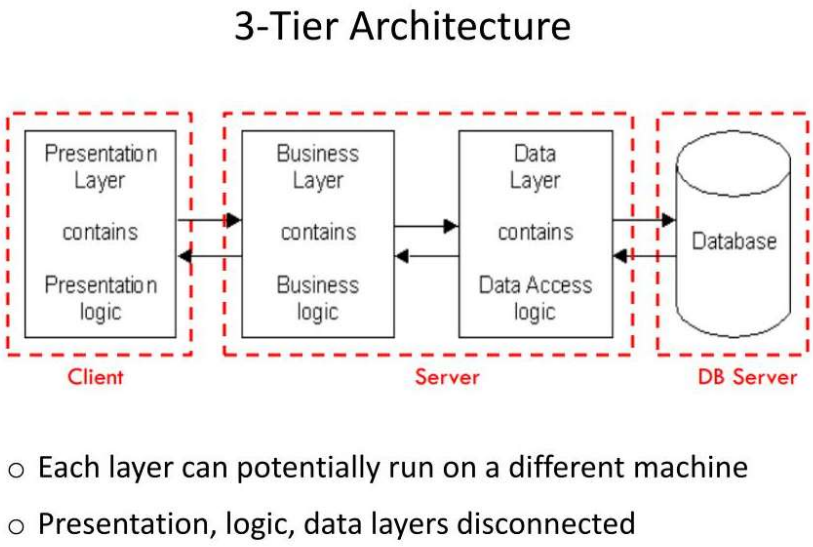


Figure 6-1 Three tier architecture

Presentation Tire

The presentation tier is the user interface and communication layer of the application, where the end user interacts with the application. Its main purpose is to display information to and collect information from the user. This top-level tier can run on a web browser, as desktop application, or a graphical user interface (GUI), for example. Web presentation tiers are

developed by using HTML, CSS, and JavaScript. Desktop applications can be written in various languages depending on the platform.

Application Tire

The application tier, also known as the logic or middle tier, serves as the application's core. Here, data gathered by the presentation tier is processed according to specific business rules. This processing may involve retrieving, comparing, or modifying information stored in the data tier. The application tier is commonly built using languages like Python, Java, PHP, or Ruby, and communicates with the data tier through API calls.

Data Tire

The data tier, often called the database tier or back-end, is where the application's information is stored and managed. This tier typically consists of a relational database management system (e.g., PostgreSQL, MySQL, Oracle, Microsoft SQL Server) or a NoSQL database server (e.g., MongoDB, Cassandra, CouchDB).

6.2. Models

6.2.1 Use Case Diagram

In the Unified Modeling Language (UML), a use case diagram can summarize the details of your system's users (also known as actors) and their interactions with the system. To build one, you'll use a set of specialized symbols and connectors.[15]

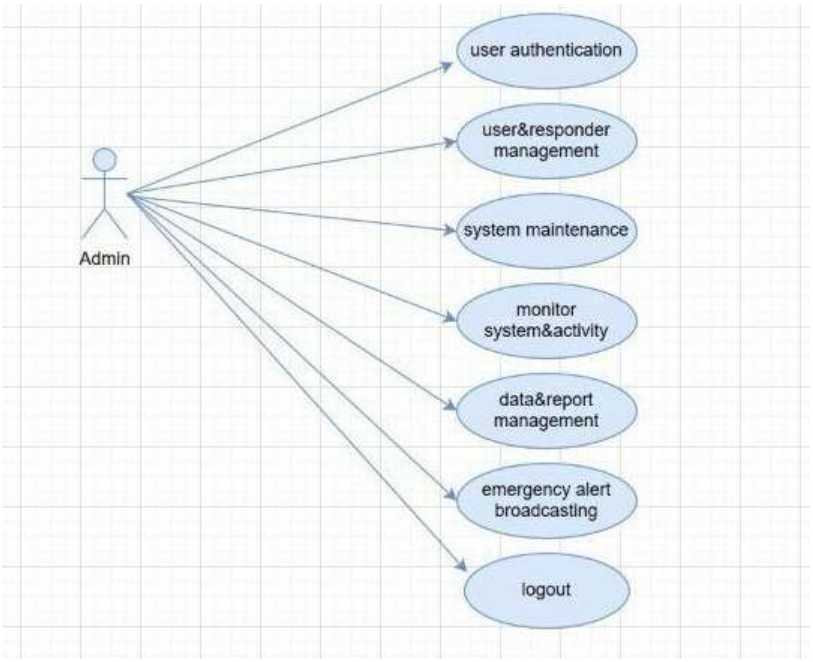


Figure 6-2 Use case diagram for admin

The Use Case diagram illustrates the basic tasks of the system admin and their management of key functions.

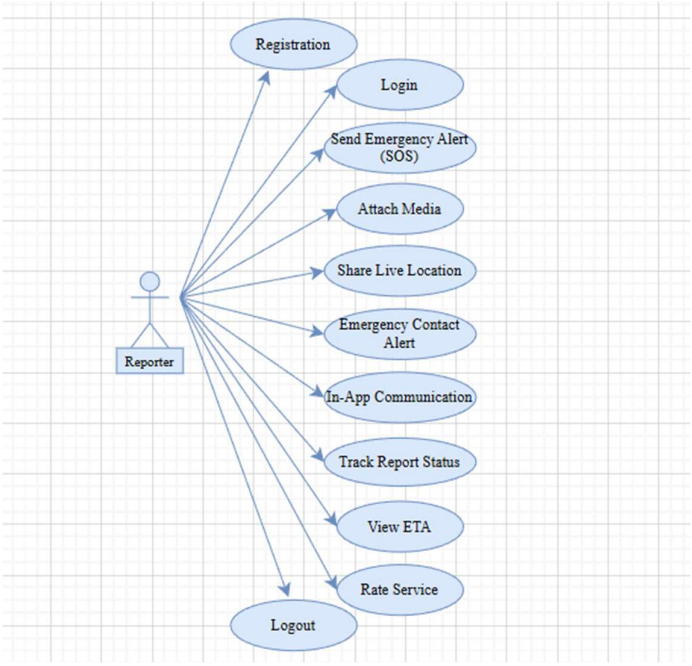


Figure 6-3 Use case diagram for Reporter

This diagram shows how the Reporter interacts with the system to report incidents and communicate during emergencies. It illustrates the Reporter's basic role in using the system.

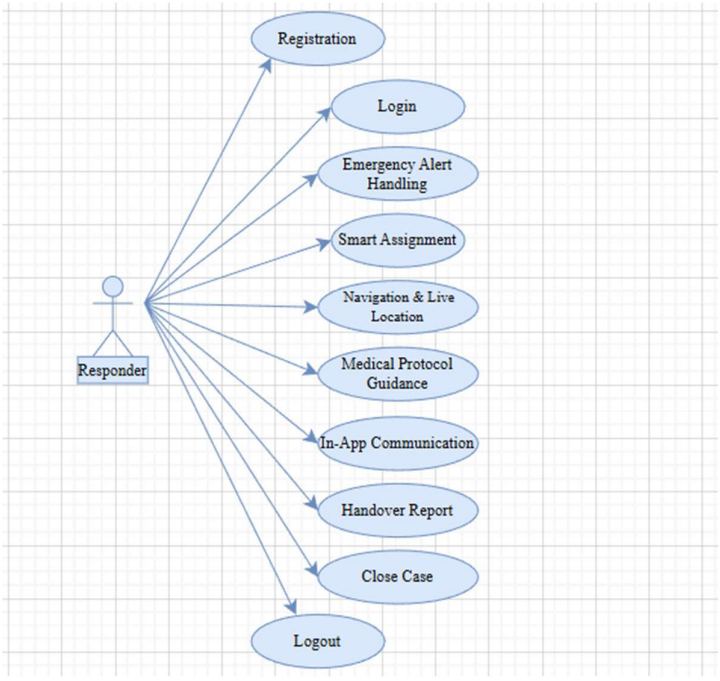


Figure 6-4 Use case diagram for Responder

This diagram shows how the Responder interacts with the system to handle reported incidents. It illustrates the Responder's role in responding to emergencies using the system.

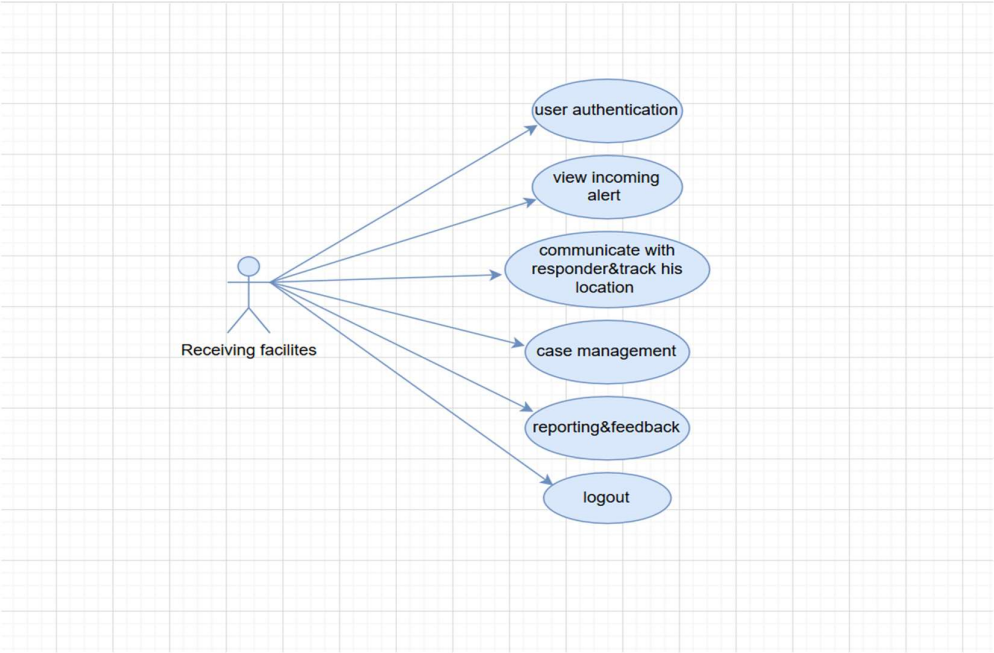


Figure 6-5 Use case diagram for receiving facilites

This Use Case diagram for the receiving facilities, highlights their main interactions with the system.

6.2.2 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation that illustrates how data moves through an information system, showing the relationship between processes, data stores, data flows, and external entities. It is one of the most used tools in systems analysis and design, helping analysts, designers, and users visualize the logic of data movement and processing without focusing on technical details such as programming or hardware.[16]

DFD Level 0 (Context Level)

A Level 0 Data Flow Diagram (DFD) also known as a Context Diagram, is a high-level graphical representation that illustrates the entire system as a single process and shows how it interacts with external entities such as users, organizations, or other systems through the flow of data. It defines the system's boundaries by depicting what data enters the system (inputs) and what data leaves the system (outputs), without showing internal processes or data stores.

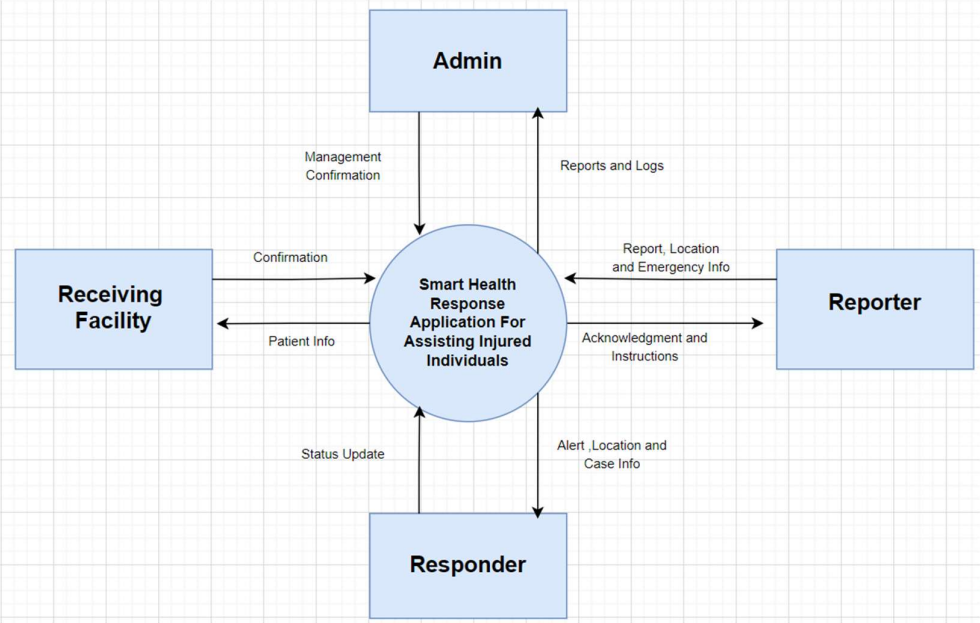


Figure 6-6 DFD Level 0 (Context Level)

This diagram shows the system as one main process and illustrates how it exchanges data with external entities such as Admin, Reporter, Responder, and Receiving Facility.

DFD Level 1

Data Flow Diagram (DFD) offers a more detailed representation of the primary system process by dividing it into multiple sub-processes. It illustrates the internal data movement between these sub-processes and highlights how the system engages with external entities like users, staff, and administrators. At this level, the diagram pinpoints data stores, outlines internal workflows, and explains the role of each system component in achieving overall functionality.

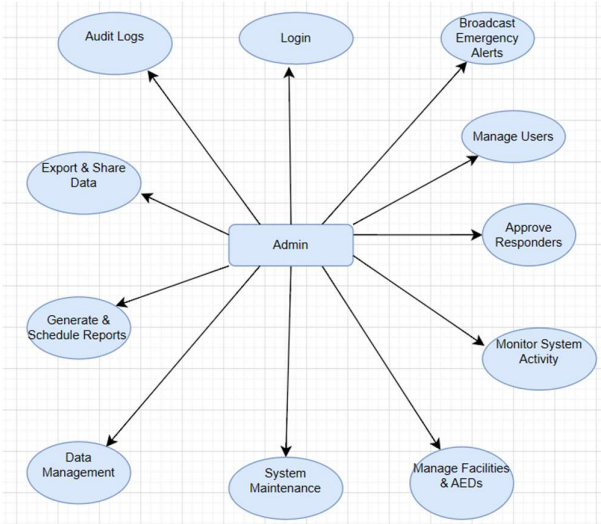


Figure 6-7 DFD-level-1 for admin

This diagram illustrates the role of the Admin user in the system. The Admin is responsible for managing users and overseeing system operations. They monitor system activity, handle data and reports, and perform system maintenance. Additionally, the Admin manages emergency alerts when needed.

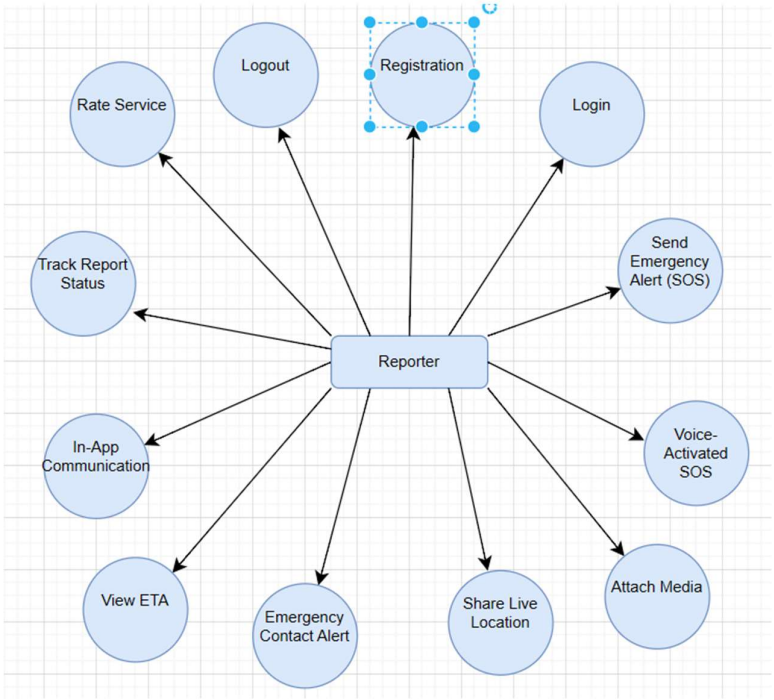


Figure 6-8-DFD level 1 for reporter

This diagram shows the role of the Reporter user. The Reporter registers and logs into the system to report emergencies. They can send emergency alerts, share their location, and attach media to support the report. The Reporter can also track the status of submitted cases.

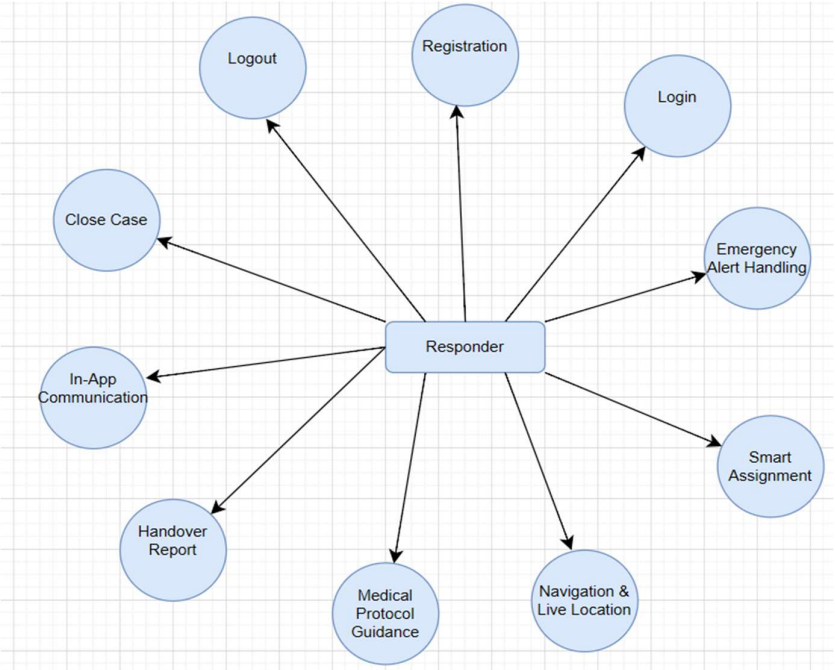


Figure 6-9 DFD level 1 for responder

This diagram presents the role of the Responder user. The Responder receives and handles emergency cases, navigates to the incident location, and follows medical guidance. After completing the response, the Responder closes the case to finalize the process.

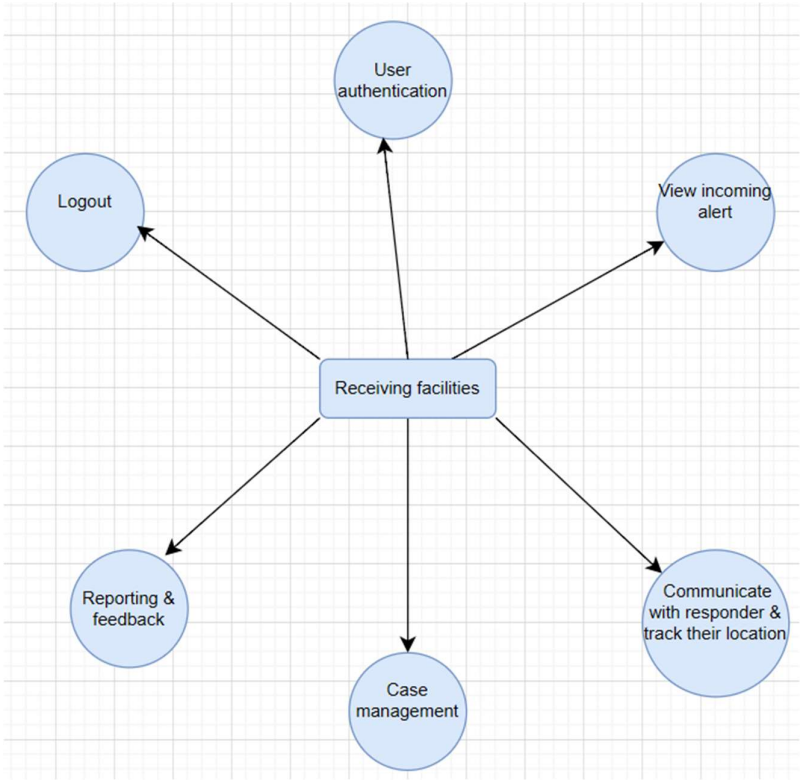


Figure 6-10 DFD-level-1 for receiving facilities

A Level 1 Data Flow Diagram (DFD) showing the main functions of receiving facilities in the system.

6.2.3 Activity diagram

An Activity Diagram is a type of UML behavioral diagram that illustrates the flow of activities within a system. It outlines how a process starts, the actions carried out, the decision points faced, and how the process concludes. This diagram provides a clear and structured representation of the sequence of steps, parallel processes, and user interactions. It is widely used to analyze system behavior, enhance workflow understanding, and confirm that all functional steps have been accurately outlined.

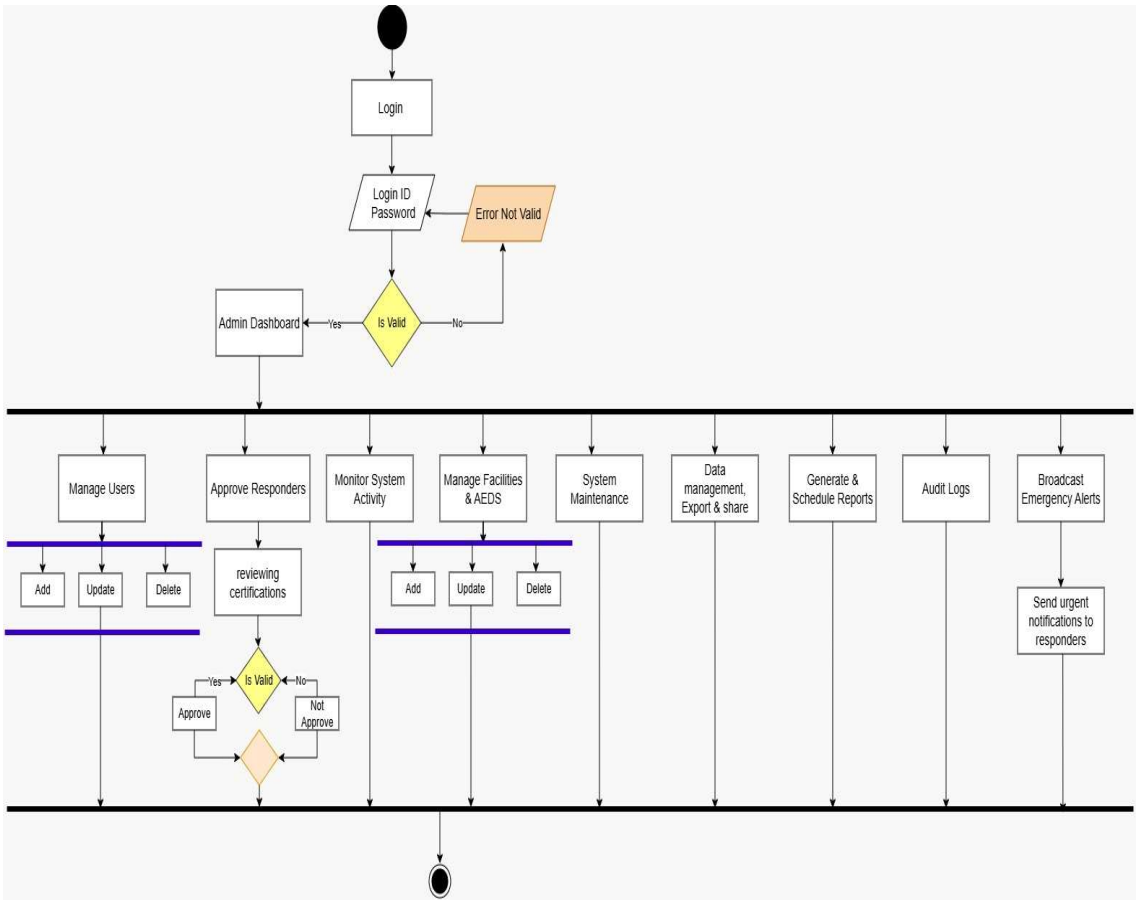


Figure 6-11 Activity Diagram for Admin

The Admin logs in to manage users, approve responders, maintain facilities, generate reports, and broadcast emergency alerts.

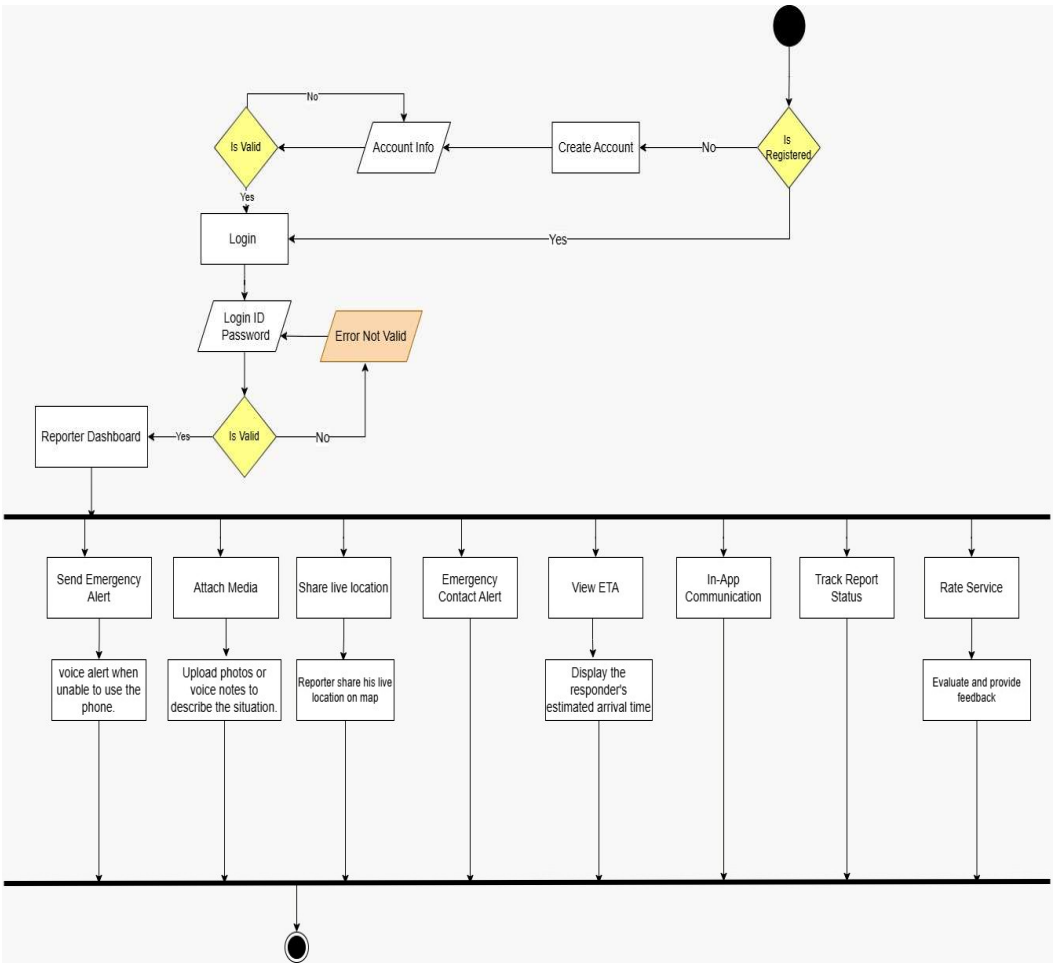


Figure 6-12 Activity Diagram for Reporter

The Reporter signs up or logs in to the dashboard to send emergency alerts, share location, attach media, track status, and rate the service.

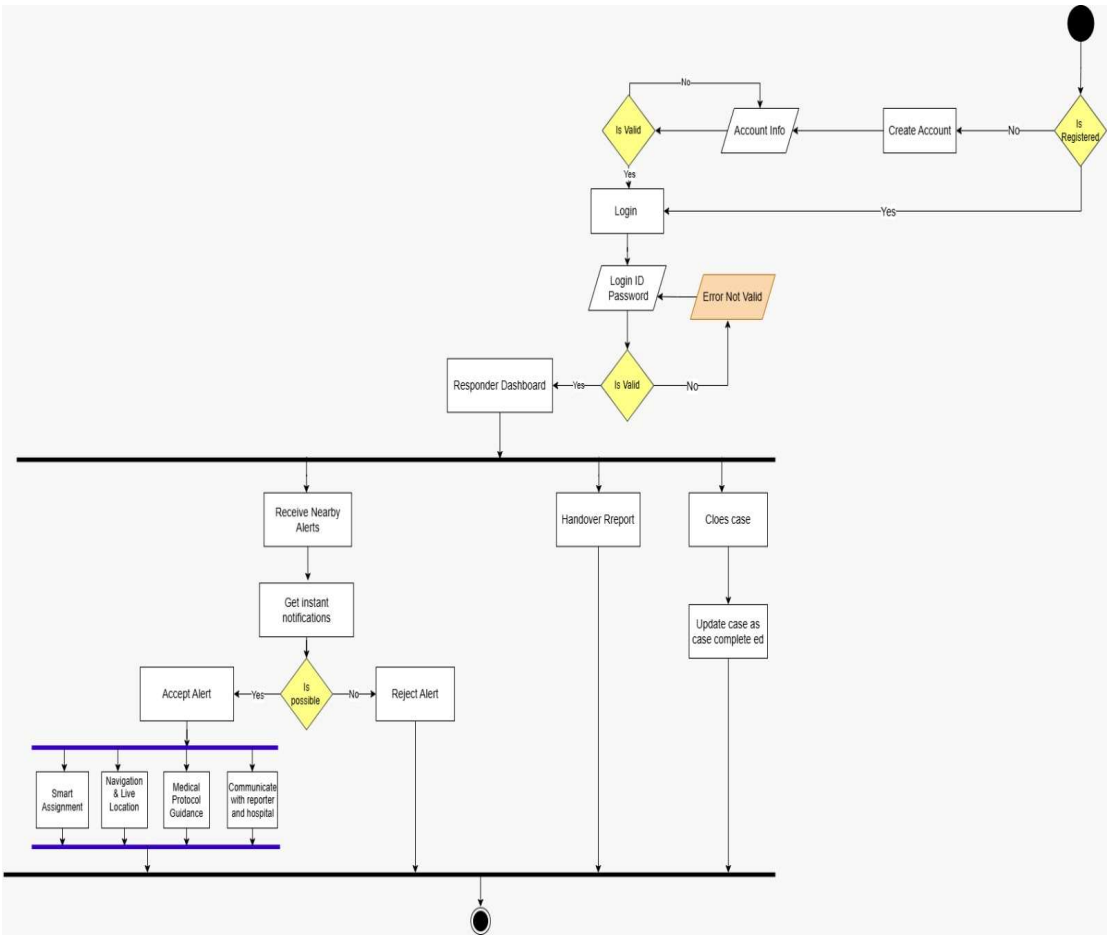


Figure 6-13 Activity Diagram for Responder

The Responder logs in to receive and accept/reject nearby alerts, navigate to the incident, and then hand over or close the case.

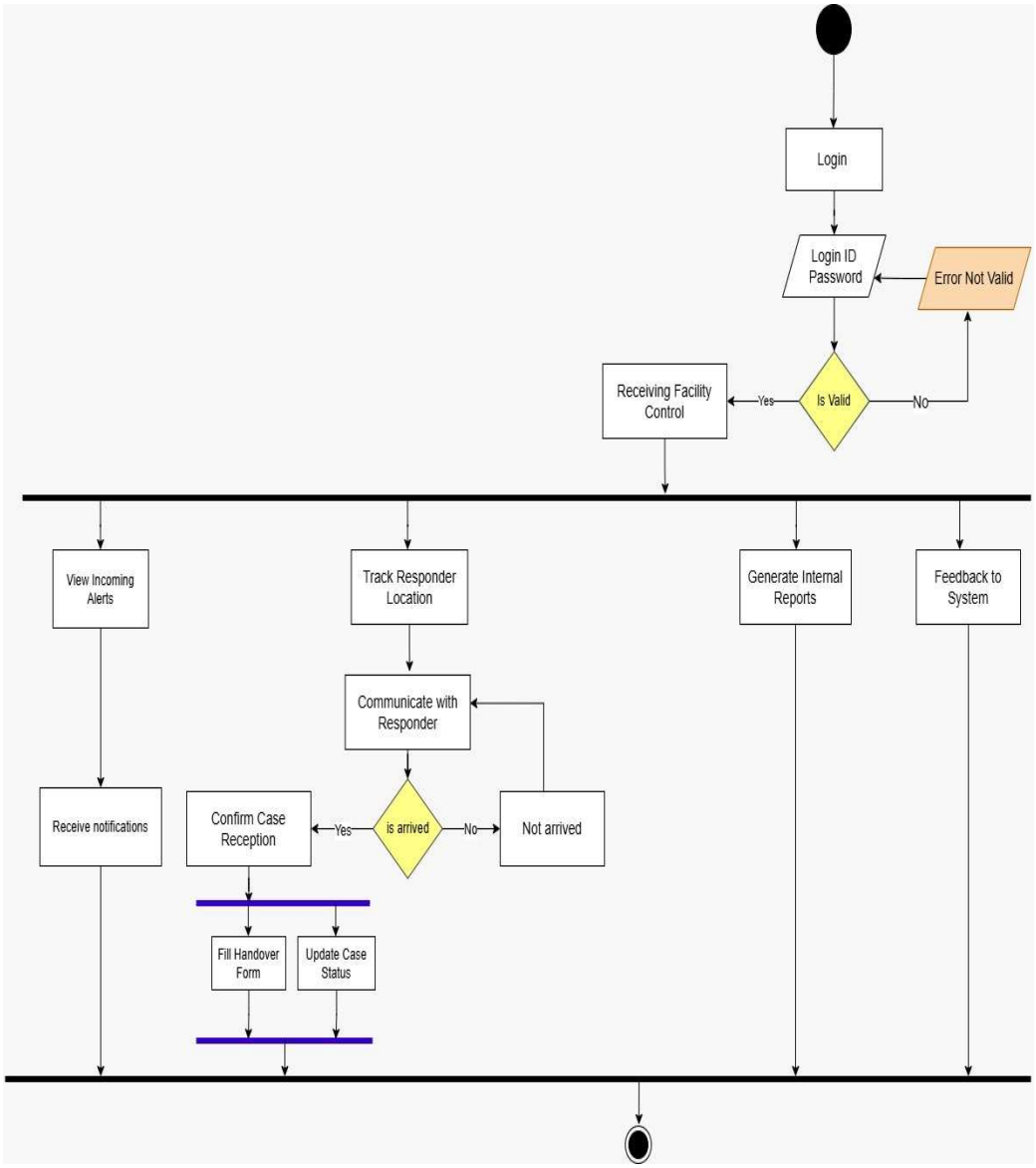


Figure 6-14 Activity Diagram for Receiving Facility

The Facility logs in to view incoming alerts, track the responder's location, communicate with them, and update the case status upon arrival.

6.2.4 Sequence Diagram

A UML sequence diagram is a graphical representation that shows how objects in a system interact over time. It captures the sequence of messages exchanged between objects and the order of these interactions, using vertical lifelines and horizontal arrows. In essence, it provides a chronological view of object interactions and offers insights into the system's dynamic behavior and flow.[17]

This visualization is essential in software and system design because it improves clarity, collaboration, and understanding. A sequence diagram serves as a visual playbook that ensures all stakeholders share a clear view of system behavior, supporting accurate development, troubleshooting, modification, and teamwork.[17]

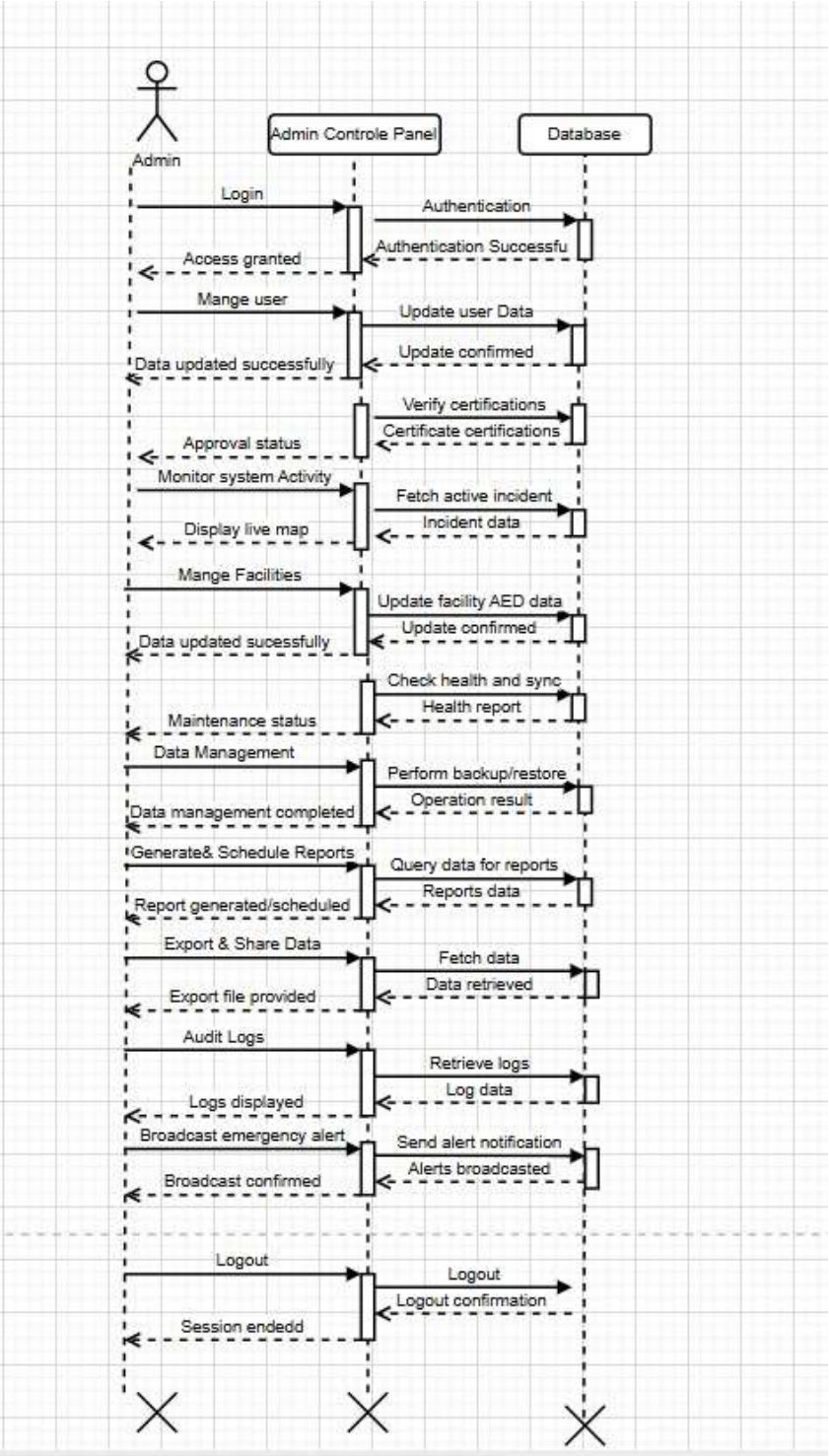


Figure 6-15 Sequence diagram for admin

A sequence diagram illustrating the interaction between the admin, the control panel, and the database for system management.

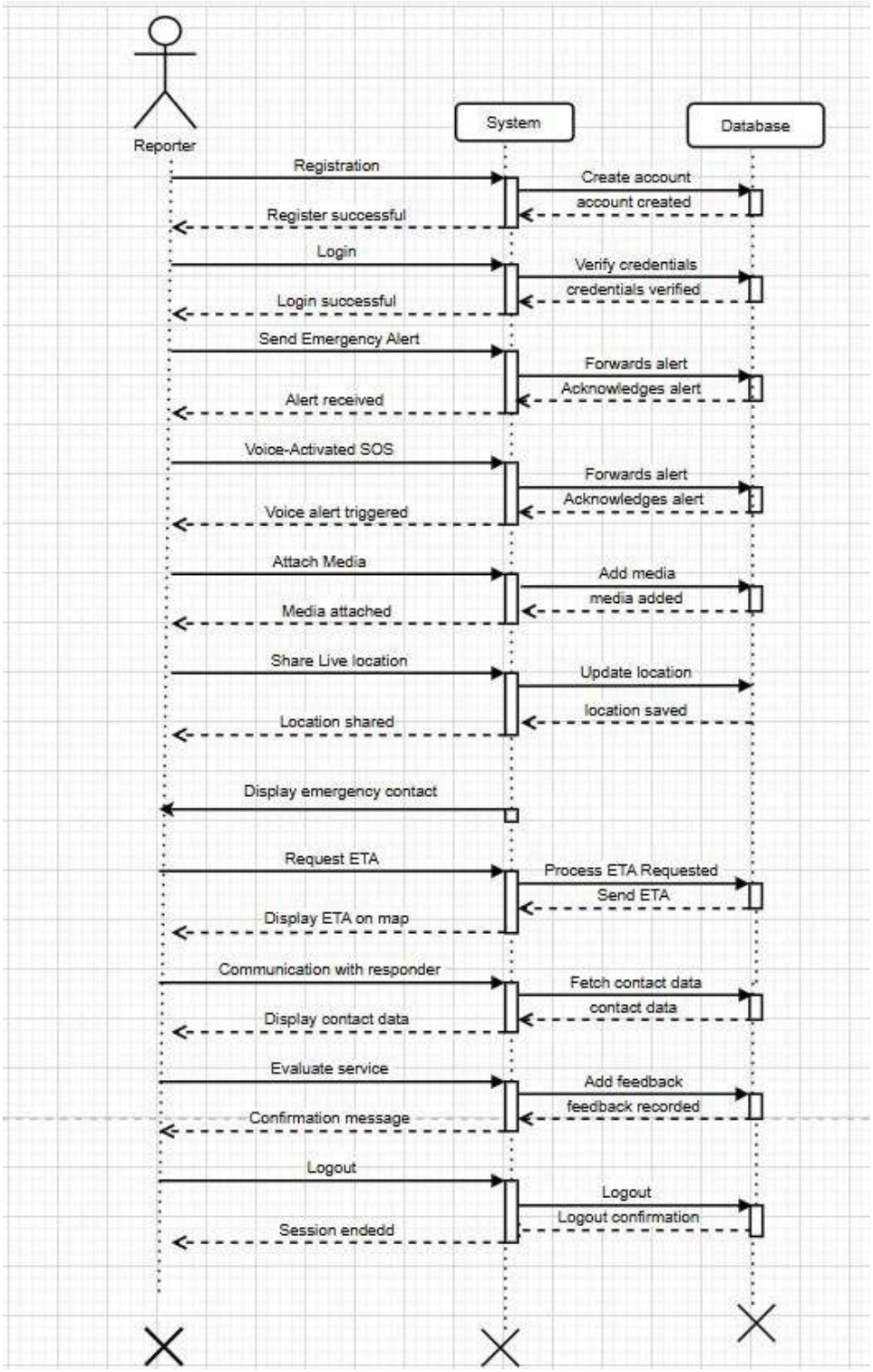


Figure 6-16 Sequence diagram for reporter

A sequence diagram showing the interaction between the reporter, the system, and the database during emergency reporting.

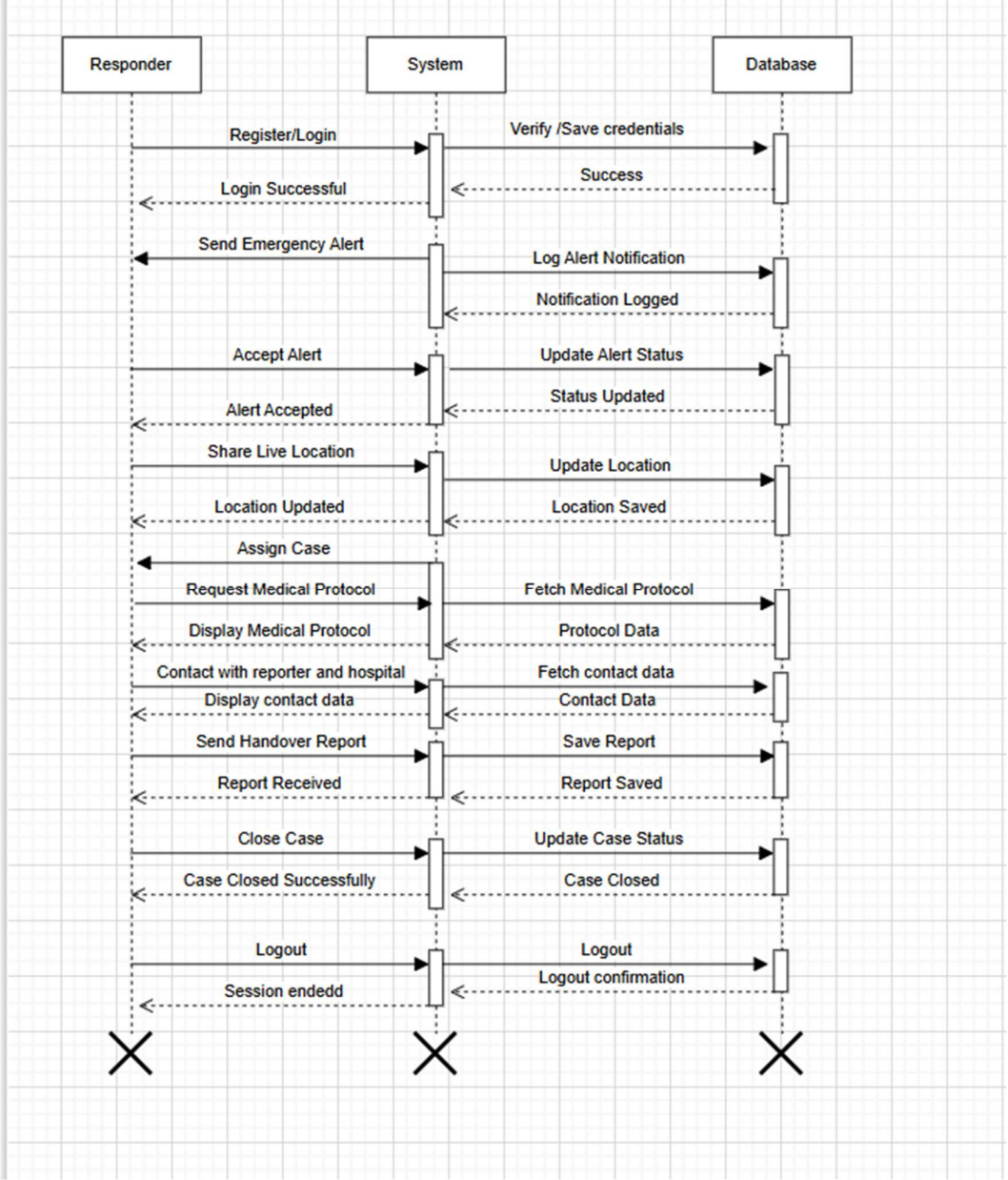


Figure 6-17 Sequence diagram for responder

This diagram Illustrates the responder's interaction with the system from login to case closure and logout.

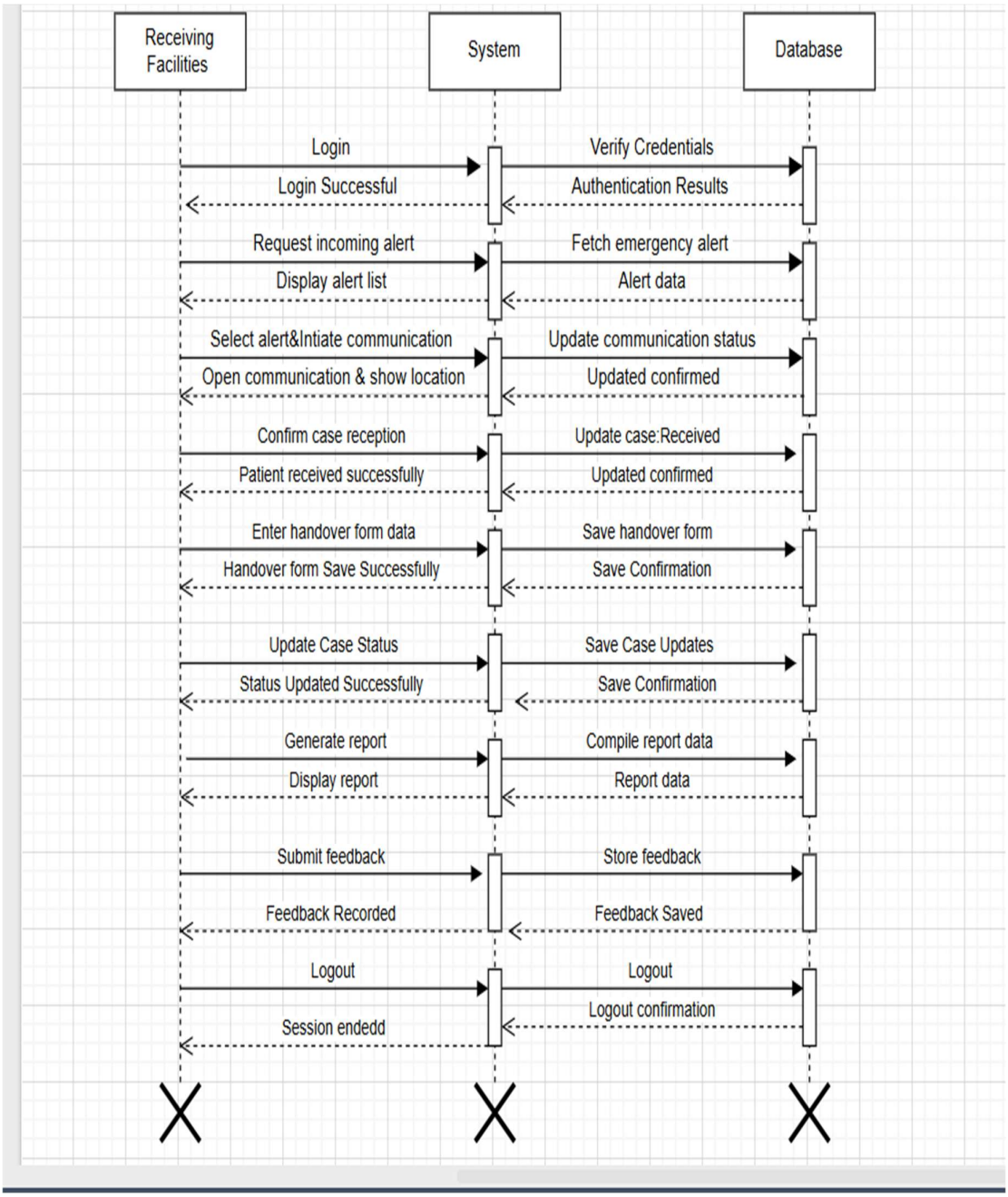


Figure 6-18 Sequence diagram for receiving facilities

This diagram Illustrates the receiver's interaction with the system from receiving the report to case closure and logout.

6.2.5 Class Diagram

A **class diagram** describes the static structure of a system, consisting of a set of classes and the relationships defined between them. A class provides the description of a group of objects and specifies their attributes and operations, thereby forming the fundamental building blocks for representing the structural aspects of a model.[18]

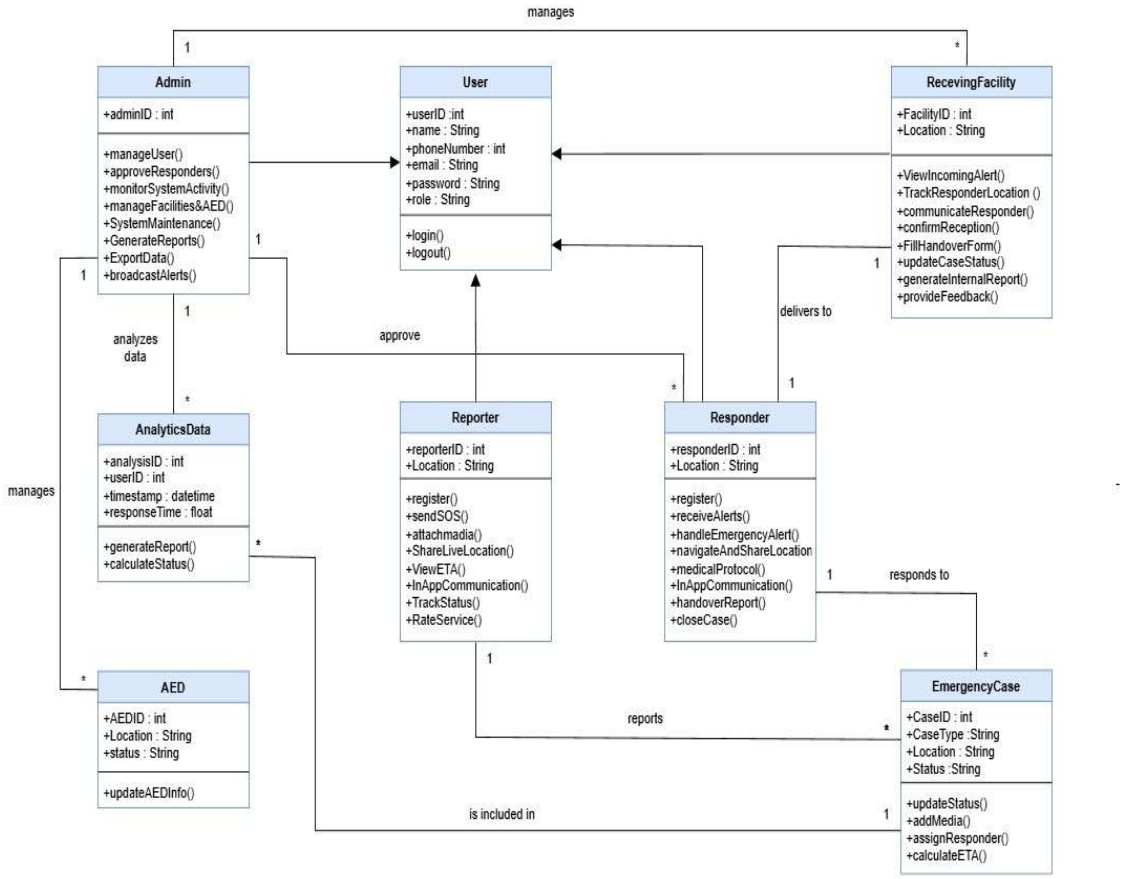


Figure 6-19 Class diagram

This class diagram illustrates the main classes of the smart health response application for assisting injured individuals, including Admin, User, Reporter, Responder, EmergencyCase, ReceivingFacility, AED, and AnalyticsData.

It shows how these classes interact with each other to support emergency reporting, response handling, system management, and data analysis.

6.2.6 Entity Relationship Diagram (ERD)

An **Entity–Relationship Diagram (ERD)** is a graphical model used during database design to illustrate how data is structured, organized, and connected within a system. It gives a high-level conceptual view of the system's entities, their attributes, and the relationships between them, which helps designers understand the system clearly before building the actual database.[19]

In an ERD:

1. **Entities:** are the object, person, event, or concept that you need to store data about (e.g., "Customer" or "Order").

2. **Attributes:** is a detail or property that describes an entity (e.g., "Customer Name" or "Order Date").

3. **Relationships:** describe how two entities are connected or interact with each other. It becomes the basis for linking tables using keys (e.g., "Places" or "Contains").

4. **Cardinality:** shows how many instances of one entity can be associated with another (One-to-One، One-to-Many, Many-to-Many).

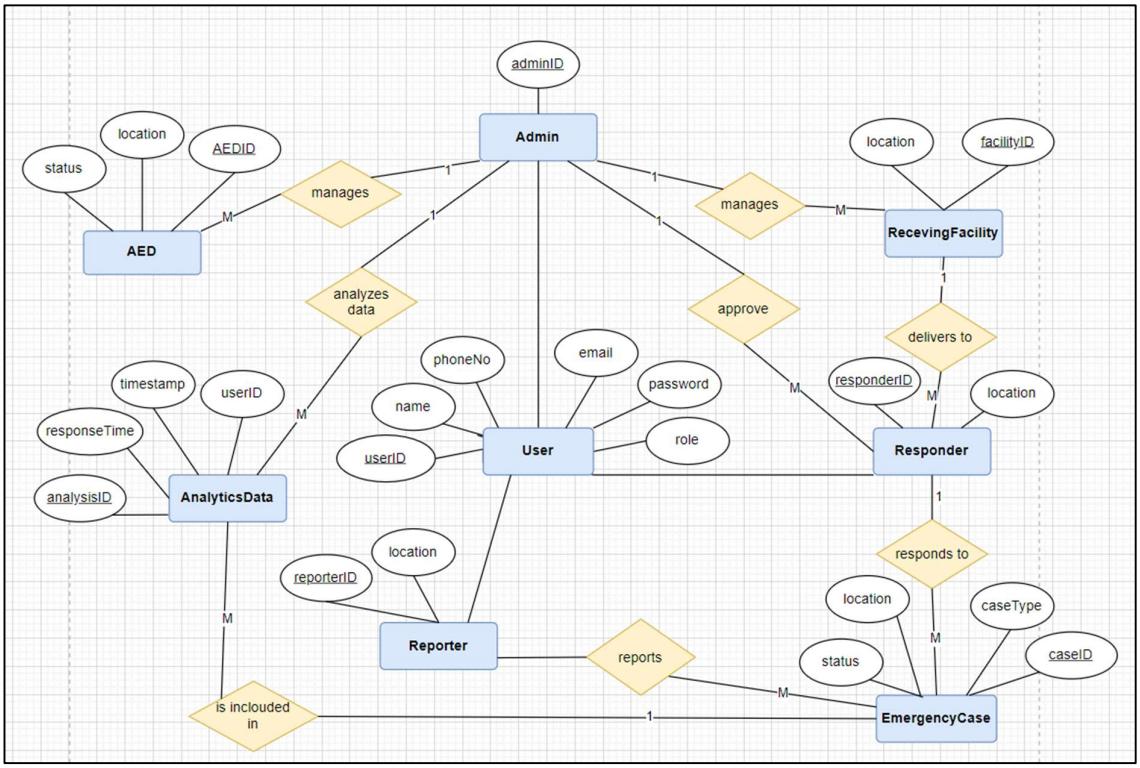


Figure 6-20 Entity relationship diagram (ERD)

This diagram illustrates the database structure by showing the main entities, their attributes, and the relationships between them.

6.3. Database Design

The database for this project was implemented using Firebase as the chosen database system. The main objective of the database is to efficiently store, retrieve, and manage data related to the emergency application while ensuring data consistency, integrity, and real-time performance. The system consists of eight tables, which are Admin, AED, Receiving Facilities, Analytics Data, User, Responder, Reporter, and Emergency Case.

6.4. Database Table overview

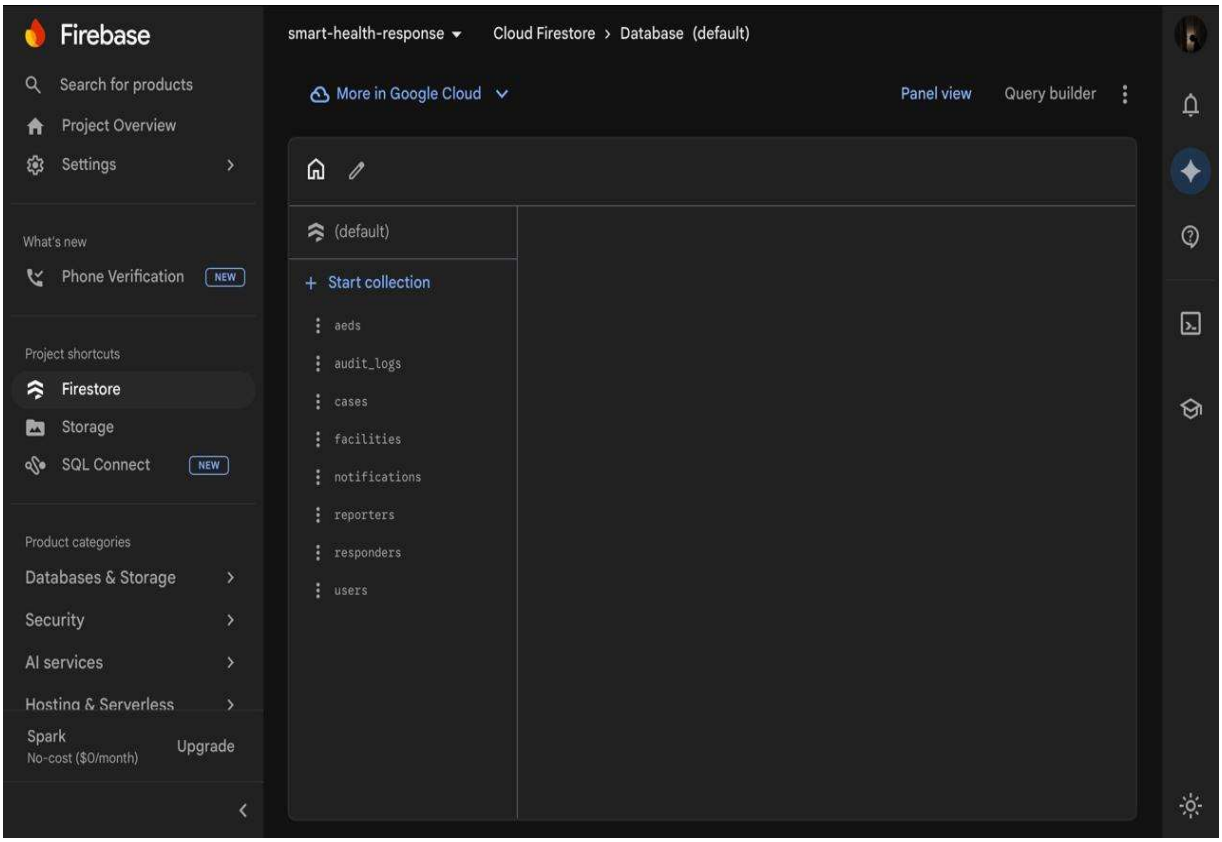


Figure 6 -1 Firebase Database Tables Over verview

6.5Audit Logs Table

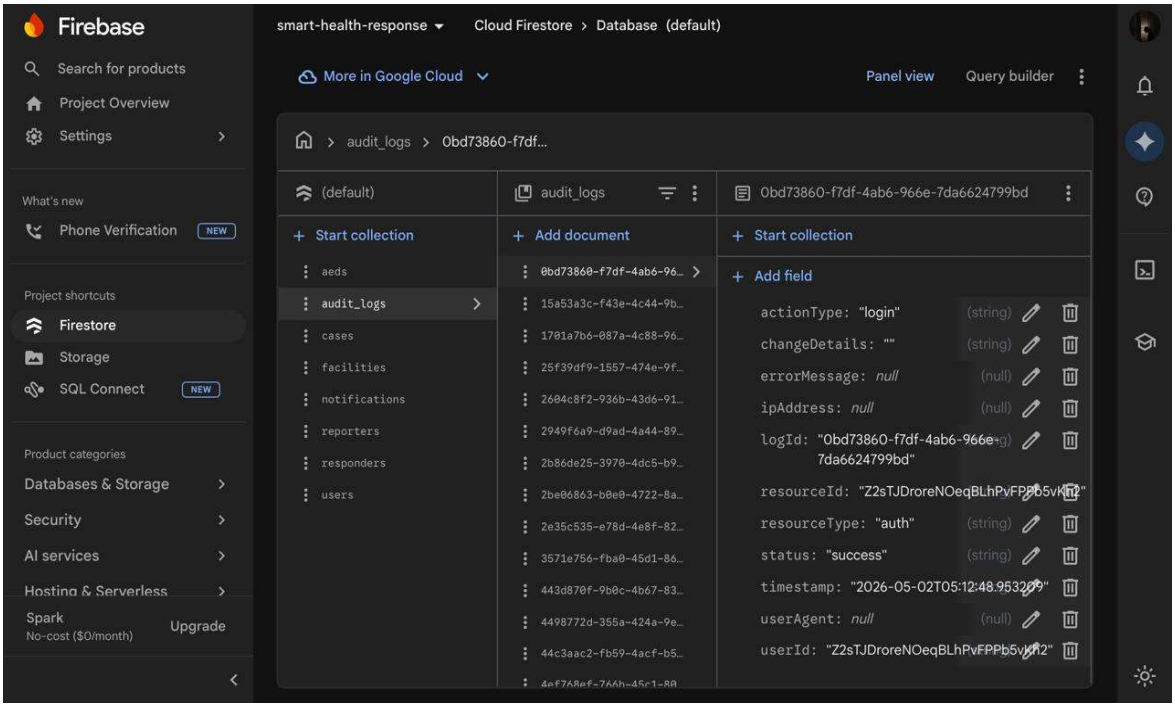


Figure 6-2 Audit Logs Table

6.6AEDs Table

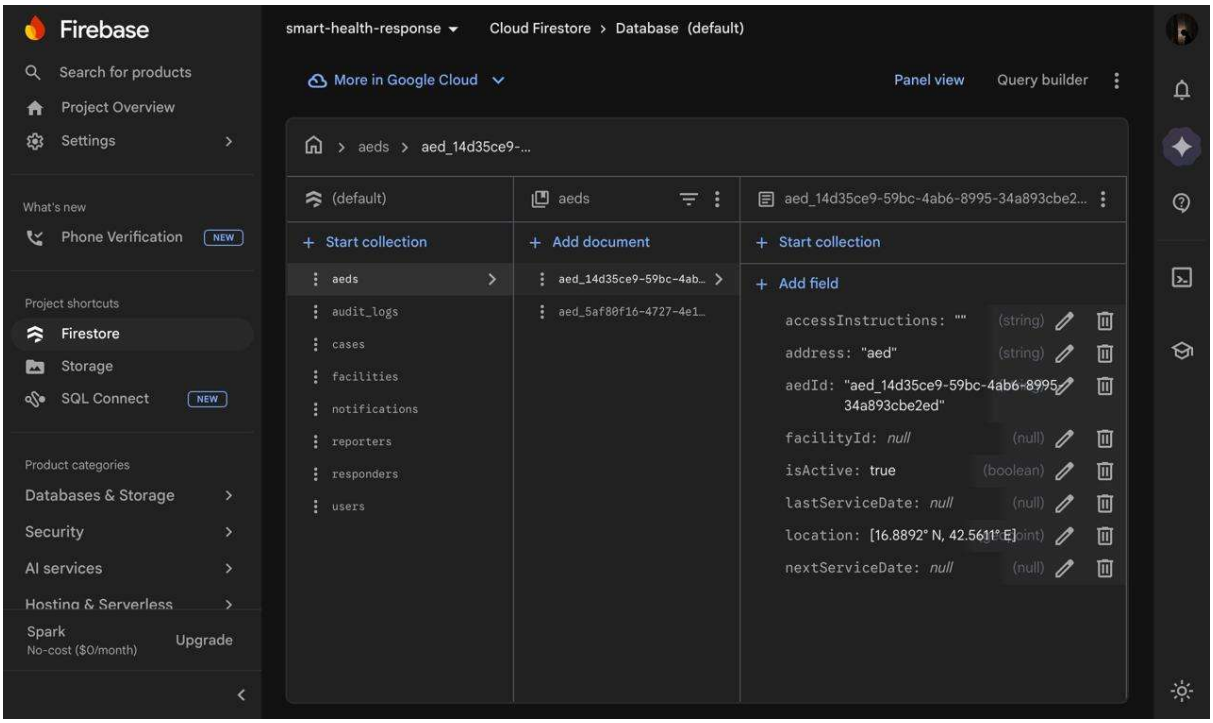


Figure 6-3. AEDs Table

6.7.Database Design

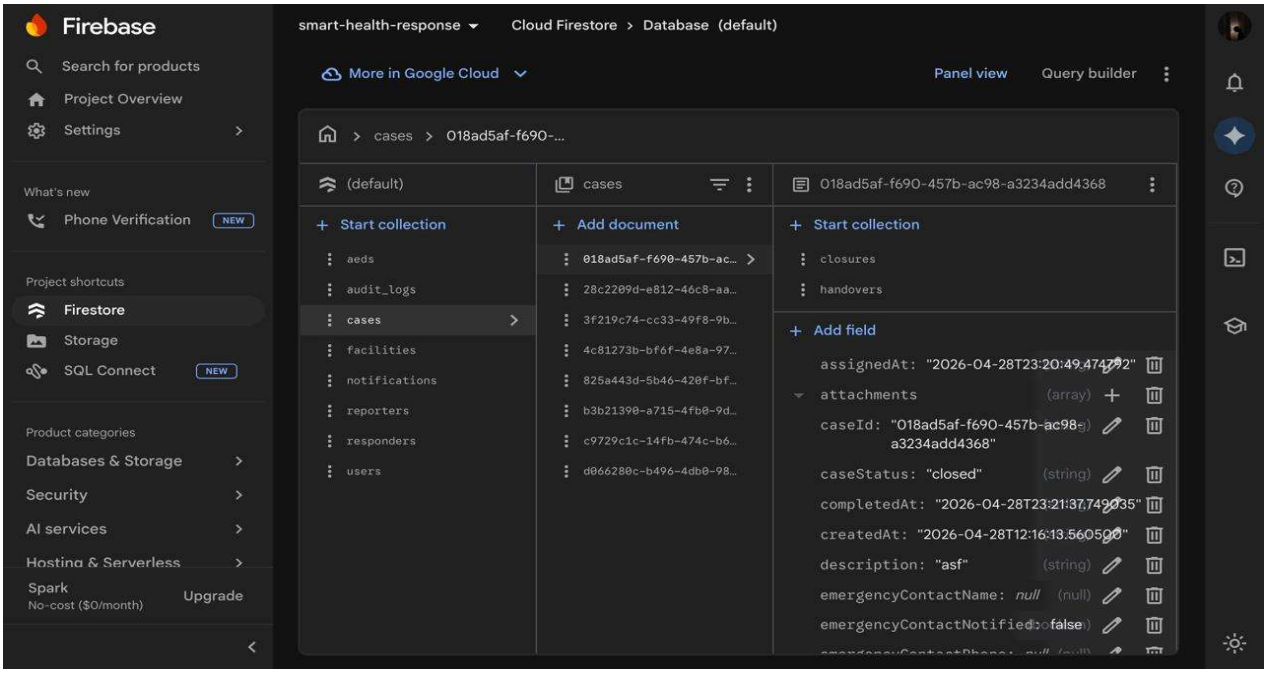


Figure 6-4 Cases Table

6.8. Facilities Table

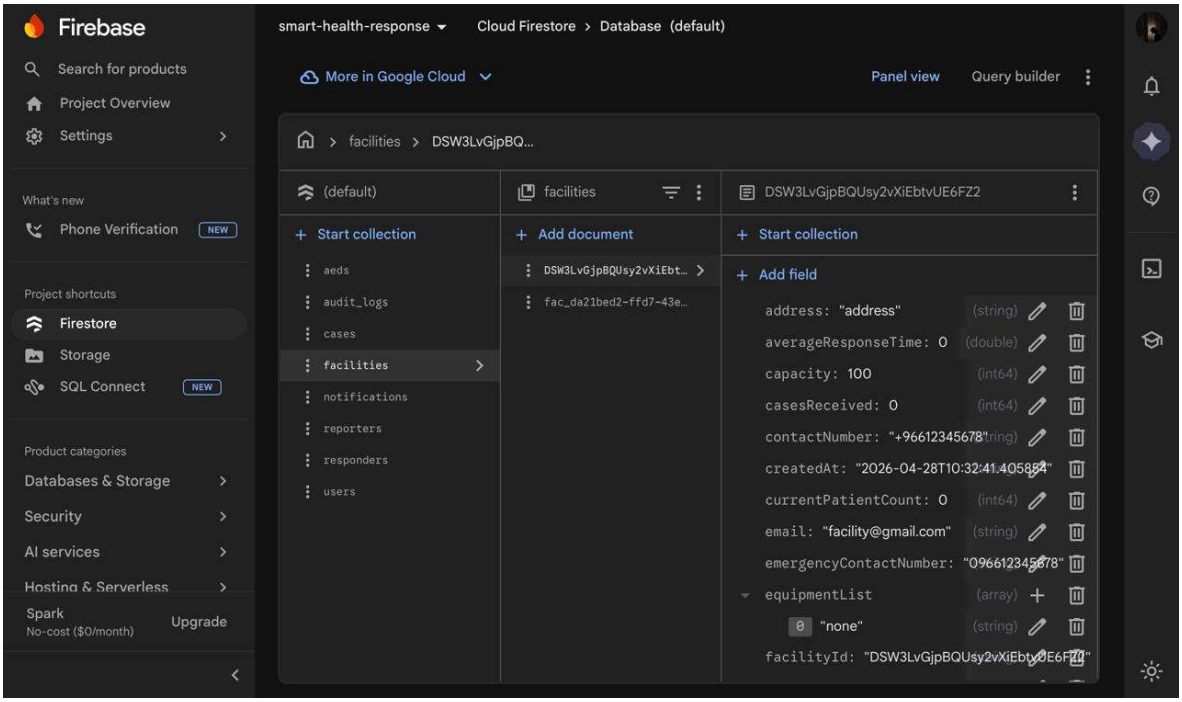


Figure 6-5 Facilities Table

6.9. Database Design

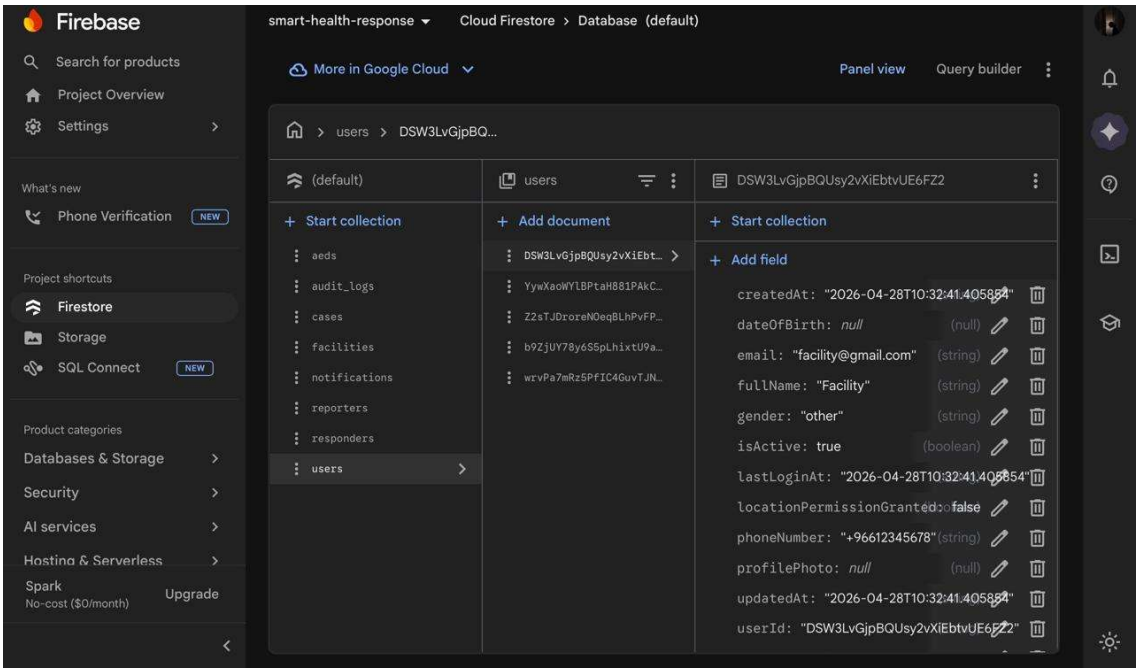


Figure 6-6 Users Table

6.10.Reporters Table

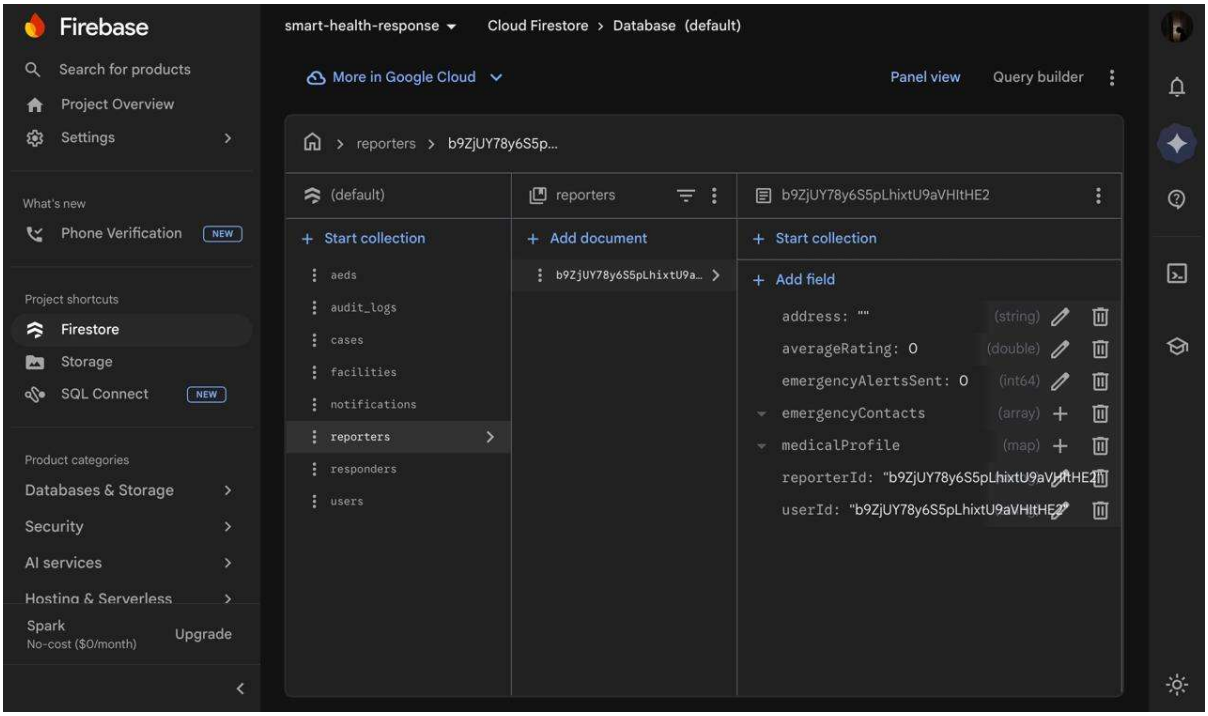


Figure 6-7 Reporters Table

6.11. Responders Table

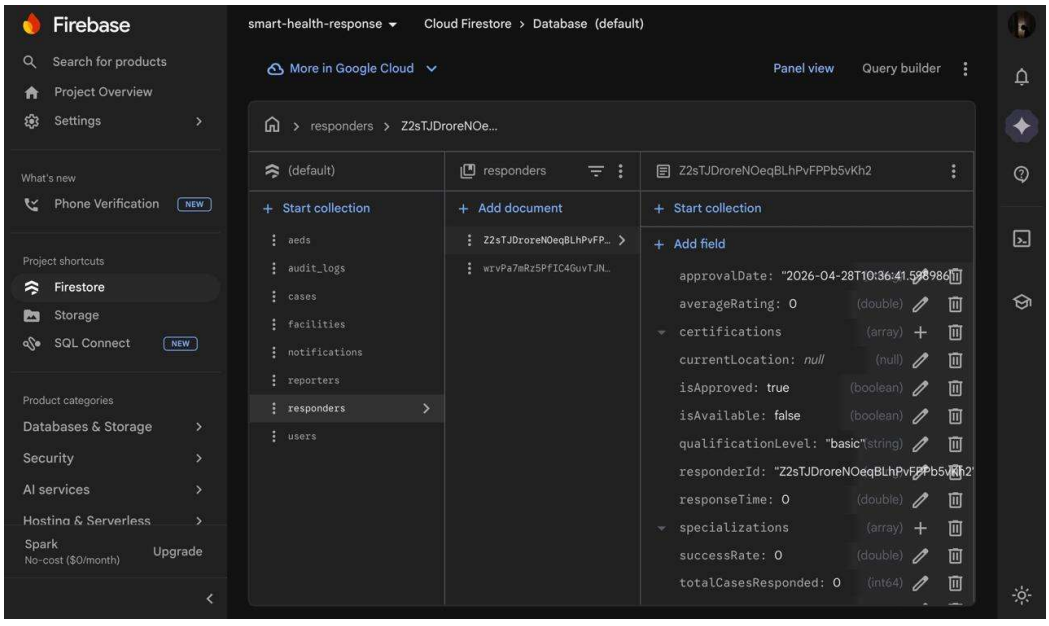


Figure 6-8 Responders Table

6.12.Notifications Table

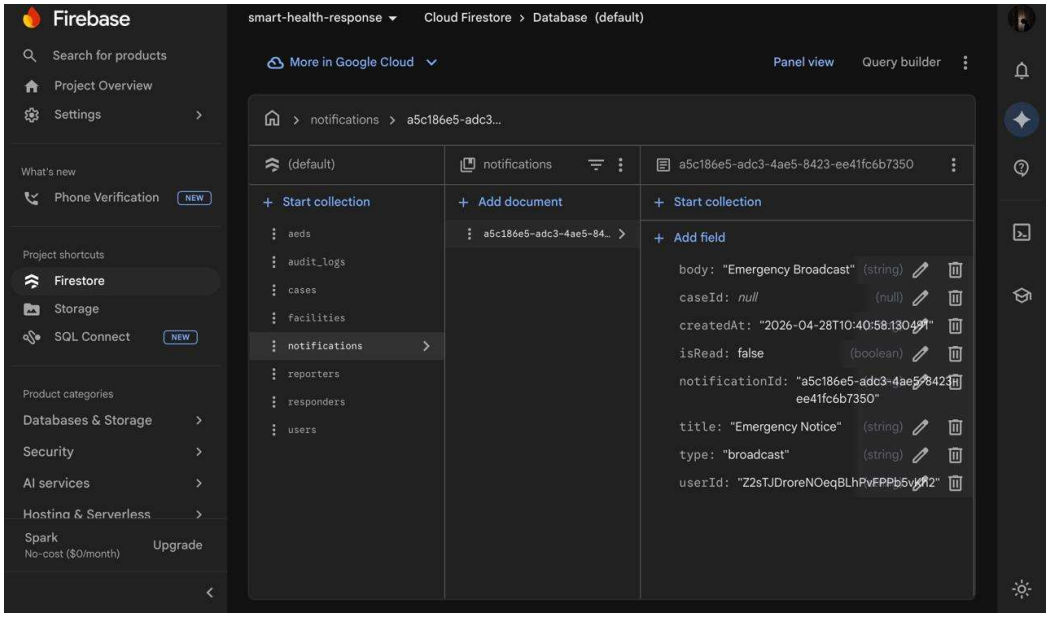


Figure 6-9 Notifications Table

7. IMPLEMENTATION OF USER INTERFACES

The user interface of the **SMART HEALTH RESPONSE APPLICATION FOR ASSISTING INJURED INDIVIDUALS** was designed to ensure simplicity, ease of use, and accessibility in addition to fast interaction and response, especially in emergency situations where time is a critical factor. The application adopts a user-friendly design that allows injured individuals or nearby users to request assistance quickly with minimal input. The interface is also designed to be easy to understand and use even under pressure, helping users take quick action and ensuring an efficient and reliable experience in critical situations.

The user interface consists of several main screens, including the emergency request screen, user registration and login, in addition to a quick access screen designed for users without an emergency account, as well as the real-time tracking view and the medical information display screen. The emergency request screen is considered the core component of the application as it provides a one-tap emergency alert feature that enables the user to send their location and condition to the system.

User interaction elements such as buttons, forms, and alerts were designed following usability principles to reduce cognitive load and ensure fast response times. Input validation mechanisms are implemented to ensure accurate data entry, particularly for medical information.

Additionally, the user interface is integrated with backend services through APIs, enabling real-time data exchange such as location tracking, status updates, and notifications. This integration directly contributes to improving response time as it ensures that information is

delivered instantly and accurately enhancing the overall efficiency of the application in handling emergency situations.

The application interfaces were designed using Figma, and the application was developed using Flutter to implement the interface design and integrate it with the Firebase database.

7.1. Main Screen

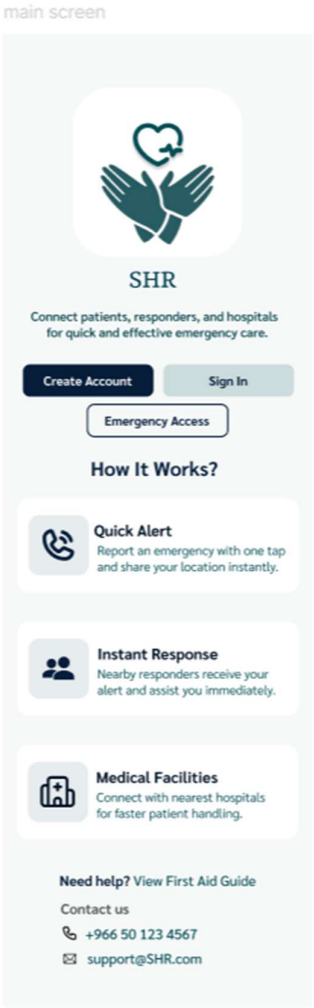


Figure 7-1 main screen

This screen represents the main entry point of the application. It allows users to create an account, sign in, or access emergency services directly through the emergency access feature. It also provides a brief overview of the application's key features to help users quickly understand the application's functionality.

7.2. Create Account Screen

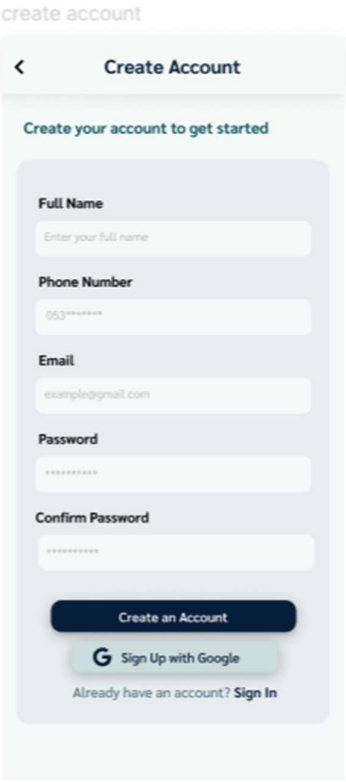


Figure 7-2 Create Account Screen

This screen allows users to quickly and easily create a new account. Users are required to enter basic information such as full name, phone number, email, and password. It also includes an option to sign up using a Google account for faster registration.

7.3. Sign In Screen

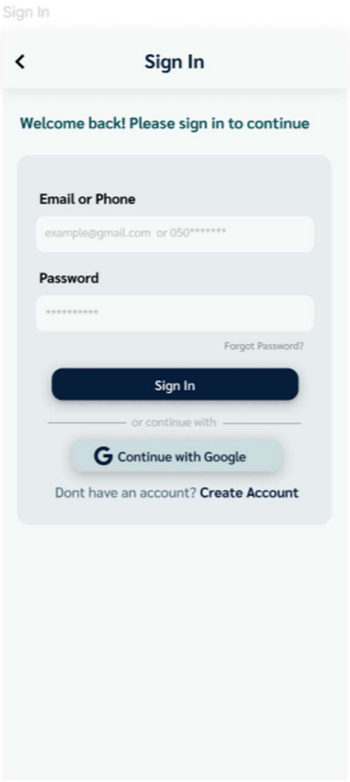


Figure 7-3 Sign In Screen

This screen allows users to access their accounts by entering their email or phone number and password. It also provides an option to sign in using a Google account for convenience.

7.4. Emergency Access Screen

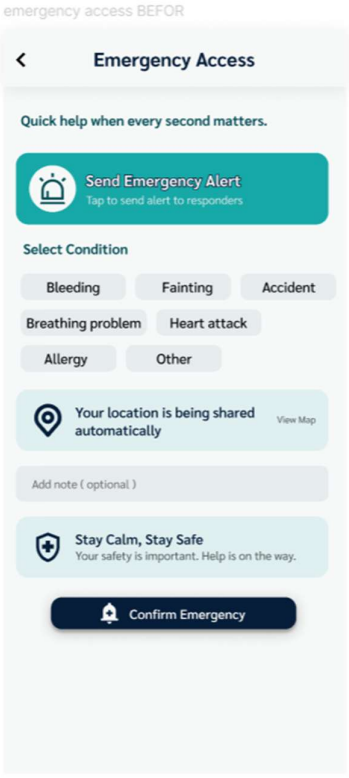


Figure 7- 4 Emergency Acess Screen

This screen allows users to quickly send an emergency alert without the need to create an account. Users can select their condition, and their location is automatically shared with the system. The screen also includes an option to add additional notes and confirm the emergency request.

Emergency Status Screen

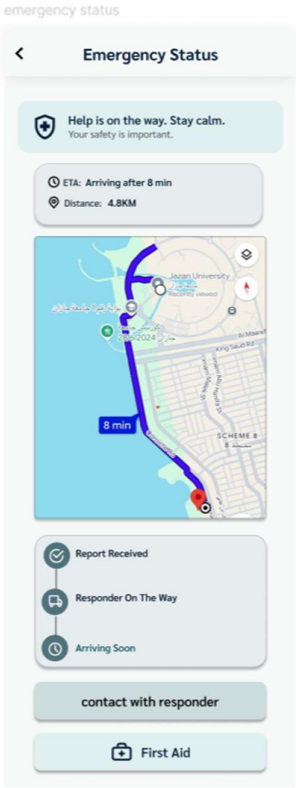


Figure 7- 5 Emergency Status Screen

This screen is displayed after an emergency alert is sent from the emergency access screen. It provides real-time updates about the status of the request, including the estimated arrival time (ETA) and the responder's location on the map. The screen also shows the progress of the request and allows the user to contact the responder or access first aid information.

Select Role Screen

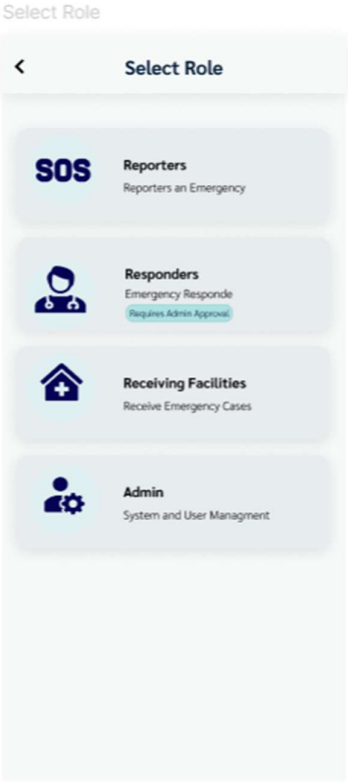


Figure 7- 6 Select Role Screen

This screen is displayed after account creation. It allows users to select their role after which they are directed to their dashboard.

Admin

The admin is responsible for overseeing the overall system and managing its operations. This includes managing users, approving responders, monitoring system activities, and generating reports and statistics, in addition to handling other administrative tasks to ensure the system operates efficiently.

7.5. Admin Dashboard Screen



Figure 7- 7 Admin Dashboard Screen

This screen provides an overview of the system through statistical information related to cases and users. It also includes a quick access section that allows the admin to navigate important pages such as reports, responder approval, and emergency alerts. Additionally, a navigation bar is available at the bottom of the screen enabling easy access to main sections such as Home, Responders, Cases, and More.

Manage Users Screen

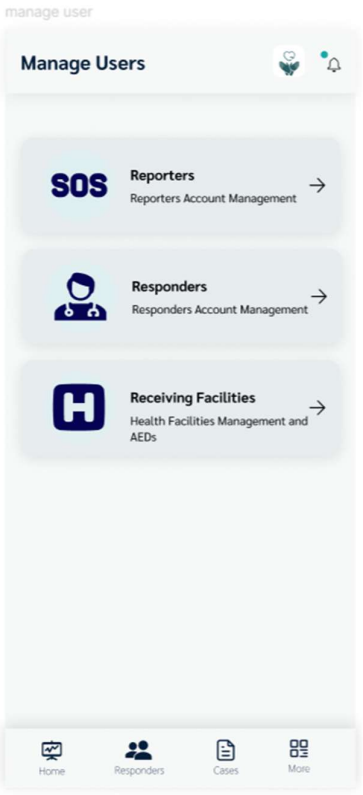


Figure 7- 8 Manage Users Screen

This screen allows the admin to manage all users in the system. The admin can view user information and perform basic actions such as adding, editing, or deleting accounts as needed.

Responder Approval Screen

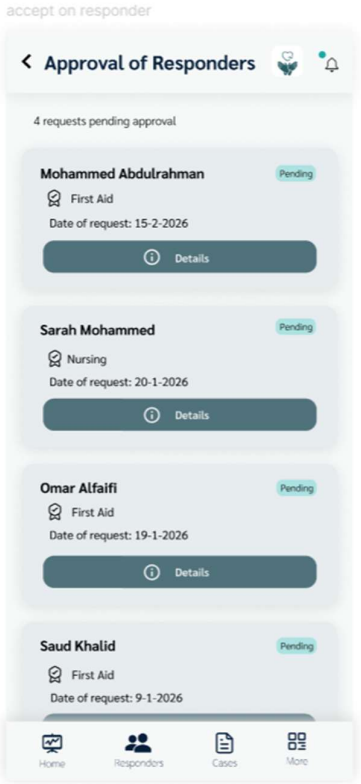


Figure 7- 9 Responder Approval Screen

This screen allows the admin to review responder requests. The admin can view information and details of each request, then decide whether to approve or reject the request to ensure only qualified responders are accepted into the system.

Manage Cases Screen

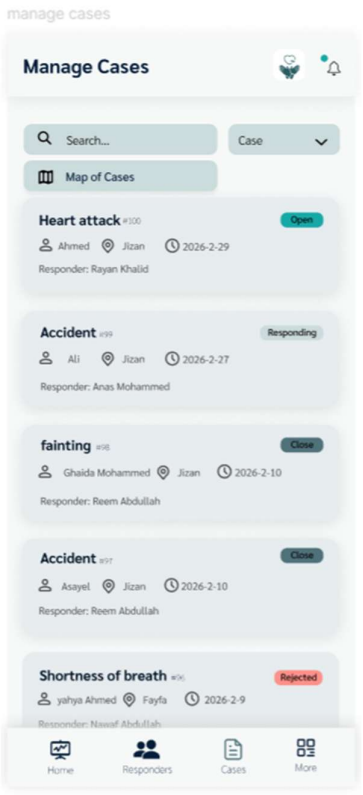


Figure 7- 10 Manage Cases Screen

This screen allows the admin to view and manage all reported cases in the system. It displays details about cases along with their status such as open, closed, or rejected, helping the admin monitor and track cases effectively.

Emergency Alerts Screen

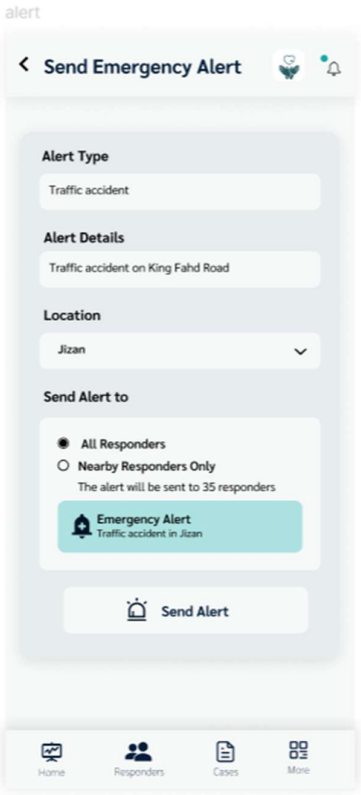


Figure 7- 11 Emergency Alerts Screen

This screen allows the admin to send emergency alerts directly to responders. It enables the admin to specify alert details and location, then notify all or nearby responders, ensuring quick response in critical situations.

Reporter

A user who reports emergencies through the application by sending alerts with location and relevant details.

7.6. Reporter Dashboard Screen

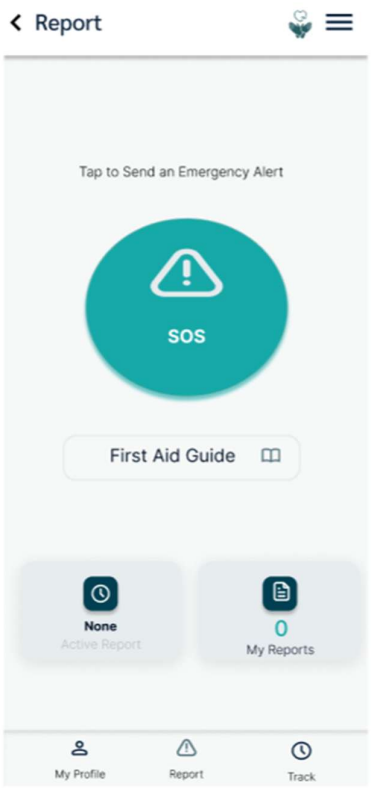


Figure 7- 12 Reporter Dashboard Screen

This screen allows the reporter to quickly send an emergency report using the SOS button. The user can specify the incident and location, then submit the request to notify responders. It also supports tracking the responder and accessing additional features such as first aid guidance and communication when needed.

Responder

A trained individual who provides first aid and responds to emergency alerts by receiving notifications, reviewing case details, and assisting the reporter.

7.7. Responder Dashboard Screen

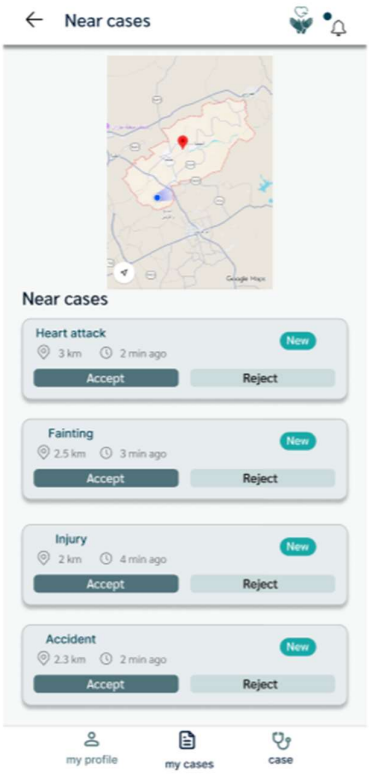


Figure 7- 13 Responder Dashboard Screen

This screen displays near cases to the responder, including details such as distance and time of the report. The responder can choose to accept or reject a case, allowing quick decision-making to handle emergencies efficiently.

Case Management Screen

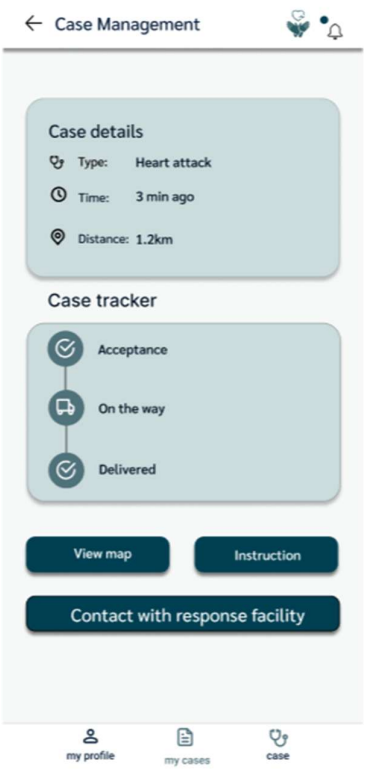


Figure 7- 14 Case Managment Screen

This screen displays case details and allows the responder to track its progress. It also includes options such as viewing the map, accessing first aid instructions, and contacting the receiving facility.

Receiving Facility

A hospital or medical center that receives emergency cases, provides appropriate medical care and confirms case completion in the system.

7.8. Receiving Facility Dashboard Screen

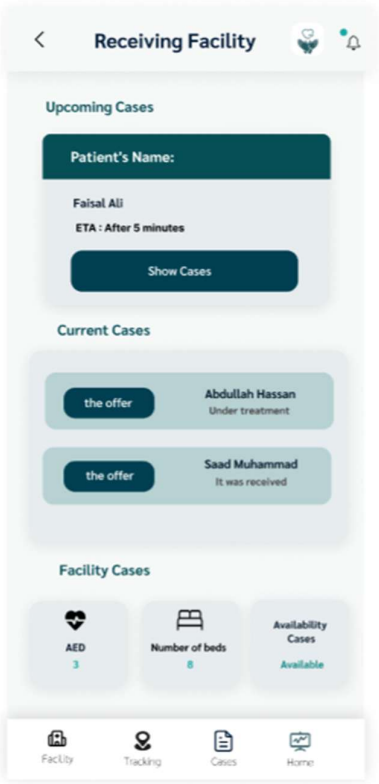


Figure 7- 15 Receiving Facility Dashboard Screen

This screen displays incoming and current cases, along with basic information about the facility's resources such as beds and equipment availability. It helps the facility manage cases and monitor its capacity.

Case Details

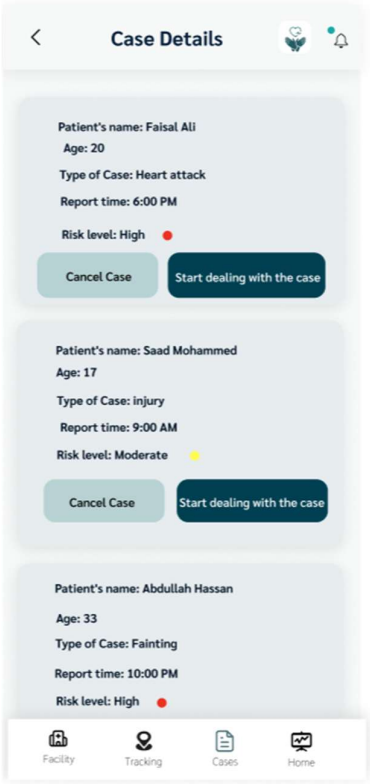


Figure 7- 16 Case Details Screen

This screen displays detailed information about incoming cases, including patient data, case type, report time, and risk level. It helps the facility assess each case and decide whether to start handling it or cancel it.

8. IMPLEMENTATION OF KEY SYSTEM MODULES

The system is composed of several key modules that work together to provide a reliable and efficient emergency response solution.

8.1. User Identity Verification Unit

This unit manages secure user registration and multi-factor authentication. Its primary role is to implement a role-based access control system.

8.2. Emergency Response Coordinator

This unit serves as the system's control center, processing incoming distress signals, analyzing the severity of situations, and issuing immediate alerts to the nearest response teams and medical facilities.

8.3. Precise Geolocation and Routing Module

This module provides continuous, real-time tracking of both the patient and the ambulance. It also calculates the fastest routes based on live traffic data to minimize response times.

8.4. Medical Data and Records Management Module

This module handles the secure storage and retrieval of organized user profiles, medical records, and incident records. It includes notifications for sending instant alerts via SMS and push notifications.

8.5. Facility Resource Optimization Unit

This unit monitors the real-time availability of hospital resources, such as bed capacity, specialized medical equipment (e.g., ventilators), and on-call specialist physicians.

8.6. Telemedicine and Multimedia Communications Unit

This unit is designed to bridge the gap between the field and the hospital, facilitating secure voice and instant messaging communication between paramedics and hospital physicians for guidance.

All modules communicate through well-defined APIs, ensuring scalability and modularity of the system. Security measures such as data encryption and secure communication protocols are implemented to protect sensitive health and location data.

9. TESTING AND VALIDATION

The Smart Health Response Application was subjected to comprehensive testing and validation processes to ensure its reliability, accuracy, and performance under emergency conditions.

Several testing approaches were employed, including unit testing, integration testing, and system testing. Unit testing was conducted to verify the functionality of individual components such as the emergency request feature, location tracking module, and notification system. Integration testing ensured that all system modules interacted correctly, particularly the communication between the user interface and backend services.

System testing was performed to evaluate the overall performance of the application in real-world scenarios. Special attention was given to response time, as the application is designed for emergency situations. The system was also tested under different network conditions to ensure consistent performance even with limited connectivity.

Validation of the application was carried out by simulating emergency cases, where users triggered distress signals and the system successfully transmitted location data and alerts to responders. The accuracy of GPS tracking and the reliability of notifications were also verified

9.1. Type of testing

Unit Testing

Unit testing focuses on verifying individual components of the application separately, such as buttons, input fields, or specific functions. The goal is to ensure that each unit works correctly on its own before integrating it with other parts of the system.

Integration Testing

Integration testing is used to evaluate how different components of the system interact with each other. It ensures that data flows correctly between modules, such as sending an emergency alert and displaying it to responders or facilities.

System Testing

System testing examines the complete application as a whole. It verifies that all features work together properly in real-world scenarios, such as handling an emergency case from alert submission to response, tracking, and closure.

No	Test type	Test name	Description	Expected result	Actual result	Status
1	Unit Test	Admin login	Test admin authentication	Admin login in successfully	Admin login in successfully	Pass
2	Unit Test	Reporter registration	Test new user registration	Account is created	Account created	Pass
3	Unit Test	Send emergency alert	Test sending emergency request	Alert is created successfully	Alert created successfully	Pass
4	Unit Test	Responder login	Test responder login	Responder Accesses system	Responder Accesses system	Pass
5	Unit Test	Facility authentication	Test facility login	Facility dashboard opens	Facility dashboard opens	Pass
6	Integration test	Alert-responder assignment	Emergency alert assigned to responder	Responder receives case	Responder receives case	Pass
7	Integration test	Location sharing-Tracking	Share live location and track it	Location updates in real time	Location updates in real time	Pass
8	Integration test	Alert-facility notification	Emergency sent to facility	Facility receives alert	Facility receives alert	pass
9	Integration test	Responder-navigation system	Navigation starts after assignment	Map open with route	Map open with route	Pass

10	**Integration test**	**Case status update- reporter view**	**Case status changes reflected**	**Reporter sees updated status**	**Reporter sees updated status**	**Pass**
11	**System test**	**Full emergency workflow**	**From alert to case closure**	**System works end-to-end**	**System works end-to-end**	**Pass**
12	**System test**	**Multi-user interaction test**	**Admin, responder, reporter, interaction**	**All role work together**	**All role work together**	**Pass**
13	**System test**	**Live Tracking scenario**	**Real-time tracking during emergency**	**Location updates correctly**	**Location updates correctly**	**Pass**
14	**System test**	**System stability test**	**General system usage**	**No crashes or errors**	**No crashes or errors**	**Pass**

Table 9-1 Type of testing

10.CONCLUSION AND FUTURE WORK

10.1. Summary of Results

This graduation project successfully designed and implemented the "Smart Health Response Application for Assisting Injured Individuals," a comprehensive mobile solution aimed at reducing emergency response times in public places within the Jazan region. The system was developed using the Waterfall methodology, ensuring a structured progression from requirements analysis through design, implementation, and testing.

The application achieved all planned objectives by delivering the following core functionalities:

- **Instant emergency reporting** with one-click SOS functionality and automatic GPS location detection
- **Role-based access control** for four distinct user types: Admin, Reporter, Responder, and Receiving Facility
- **Real-time location tracking** and navigation for responders to reach incident locations efficiently
- **First aid guidance** delivered through text, images, and video instructions for non-specialized users
- **Secure communication channels** enabling chat and calls between reporters, responders, and medical facilities
- **Comprehensive administrative dashboard** for user management, responder approval, case monitoring, and report generation

The testing and validation phase employed unit testing, integration testing, and system testing. As documented in Table 9-1, all fourteen test cases passed successfully, including critical scenarios such as emergency alert transmission (test case 1), responder navigation (test case 9), full emergency workflow from alert to case closure (test case 11), multi-user interaction (test case 12), live tracking (test case 13), and system stability (test case 14). The system demonstrated acceptable response times under varying network conditions and maintained reliable notification delivery.

The proposed system directly addresses the identified research gap by providing an accessible, easy-to-use emergency response platform that does not rely on limited

responders within specific environments. Unlike existing applications such as ASAFNY (which suffers from technical registration issues) and SAFE (limited to Android users in the Philippines), this solution offers cross-platform support (Android and iOS), automatic location sharing, offline first aid guidance, and seamless integration between all emergency stakeholders.

10.2. Limitations

Despite the successful implementation and testing of the proposed system, several limitations must be acknowledged within the context of this academic project:

Geographical Scope Constraint: The system was designed and validated specifically for the Jazan region, as stated in the project scope (Section 1.3). Consequently, the application's performance in other regions with different infrastructure characteristics, population densities, or emergency service protocols has not been evaluated.

Internet Connectivity Dependency: As noted in Section 1.4 (Limitations and Constraints), the application requires a stable internet connection to transmit emergency alerts and share real-time locations. In areas with weak or no network coverage—particularly remote or mountainous areas within the Jazan region—system performance may be degraded or unavailable. While offline first aid guidance is provided, real-time emergency coordination depends on connectivity.

Device and Power Limitations: The application depends on smartphones running Android 14 or higher, or iOS 17 or higher, as specified in the hardware requirements (Table 5-1). Users with older or incompatible devices cannot benefit from the system. Additionally, continuous GPS usage and push notifications consume significant battery power, which may affect application availability during prolonged emergencies.

Dependence on External Responders: The system can notify medical authorities, trained volunteers, and receiving facilities, but the actual response time remains contingent upon the availability, readiness, and geographic distribution of emergency services. The application cannot guarantee immediate physical response if no qualified responders are nearby or available.

Data Privacy and Security Considerations: Although the system implements user authentication, role-based access control, and data encryption, it has not undergone formal security auditing or penetration testing. Handling sensitive user data—including

real-time location, medical history, and personal identification information—requires ongoing vigilance and potential compliance with healthcare data protection regulations.

Limited User Acceptance Testing Scale: The validation process was conducted using controlled test scenarios and a limited number of simulated users rather than full-scale deployment with a large, diverse user population over an extended period. Real-world usability issues or unexpected edge cases may emerge only during prolonged operational use.

No Integration with Official Emergency Infrastructure: The current implementation does not include direct API integration with official systems such as the Saudi Red Crescent dispatching platform, Ministry of Health databases, or national ambulance coordination centers. Such integration would be necessary for full operational deployment.

10.3. Future Enhancements

Building upon the current implementation, several practical improvements and advanced features are proposed to enhance the system's effectiveness, usability, and scalability.

Near-Term Enhancements (Next Iteration)

Offline Mode Expansion: While offline first aid guidance is currently supported, future iterations should enable offline emergency report queuing, where SOS requests are automatically transmitted when connectivity is restored. This would address the limitations identified in areas with intermittent network coverage.

Multi-Language Support: Following the recommendation from the literature review of similar applications (My Nearest Aider), the application should support additional languages beyond Arabic and English, including Urdu and Bengali, to serve the diverse expatriate population in the Jazan region.

SMS Fallback Notification System: To mitigate internet dependency, an SMS-based notification system should be implemented as a secondary communication channel. When an emergency alert is triggered and internet connectivity is unavailable or unstable, the system would automatically send location and case information via SMS to the nearest registered responders and emergency contacts.

Enhanced Battery Optimization: Implement adaptive GPS sampling rates and intelligent background processing to reduce battery consumption while maintaining location accuracy during emergencies. This could include switching to lower-power location providers when the user is stationary.

Integration with National Emergency Numbers: Establish direct integration with the Saudi Red Crescent (ASAFNY) and the unified ambulance service (997) to enable automatic dispatch coordination. This would transform the application from an informational platform into an officially recognized emergency reporting channel.

Wearable Device Support: Extend application functionality to smartwatches (Apple Watch, Wear OS) to enable one-tap SOS alerts, fall detection, and heart rate monitoring without requiring smartphone interaction. This is particularly valuable for elderly users or individuals with mobility limitations.

Long-Term Future Work

Artificial Intelligence for Triage and Resource Allocation: Implement machine learning algorithms to analyze reported symptoms, vital signs (when available from wearables), and historical case data to automatically assign severity scores and prioritize the most critical cases. The system could also predict required hospital resources (e.g., ICU beds, specific specialists) before the patient arrives.

Predictive Incident Mapping: Leverage historical incident data, weather conditions, time-of-day patterns, and public event schedules to predict high-risk locations and times. The system could proactively suggest increased responder presence in predicted hotspot areas and send preventive safety alerts to users.

Drone and Automated External Defibrillator (AED) Integration: Integrate with drone delivery systems to transport AEDs, emergency medications, or communication devices to incident locations before ground responders arrive. The database schema already includes an AED entity (as shown in Figure 6-20, Entity Relationship Diagram), providing a foundation for this enhancement.

Blockchain-Based Medical Record Sharing: Implement a decentralized, permissioned blockchain system for secure, tamper-proof storage and sharing of user medical records, certifications, and incident reports. This would enhance data integrity, patient privacy, and cross-institutional interoperability.

Geographical Expansion and Regional Adaptation: Extend the system to cover the entire Kingdom of Saudi Arabia, followed by other Gulf Cooperation Council (GCC) countries. Each expansion would require adaptation to local emergency protocols, language variations, and regulatory compliance. A modular configuration system is recommended to support region-specific rules without code modification.

Advanced Analytics Dashboard for Public Health Authorities: Develop comprehensive analytics capabilities for the Ministry of Health and Red Crescent, including real-time incident heat maps, response time dashboards, resource utilization reports, and trend analysis to inform public health policy and emergency resource planning.

Voice-Activated Emergency Reporting with Natural Language Processing: Enhance the existing voice-activated SOS feature (mentioned in Section 5.2.1) with advanced natural language understanding to automatically extract case type, severity, and victim condition from unstructured speech, reducing the need for manual input during high-stress situations.

Community Training and Certification Module: Integrate a built-in training platform offering certified first aid courses, CPR training with video assessment, and public awareness content. Users who complete training could be recognized as "community verified" responders, expanding the available responder network while maintaining quality standards.

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